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Innovation Prizes as Fundraising Signals*
Evidence from XPRIZE competitions

Francois Bares† Thomas Hellmann† Elaine Hungenberg‡ Andrew Tauhert‡

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Abstract

While a prior literature established that patents have signalling value to early-stage investors, little is known about the signalling value of innovation prizes. This paper analyses PitchBook fundraising outcomes of 383 startups that participated in innovation prizes offered by the XPRIZE Foundation, the largest organiser of innovation prizes in the world. Using a synthetic difference-in-differences estimator with a large control group, the analysis finds that XPRIZE participation increases the probability of raising equity in any quarter by 7.7 percentage points, relative to a baseline of 4.7%. Consistent with signalling theory, the effect is twice as large for companies without patents. It is positive and significant in each individual prize, but larger for prizes with larger prize purses. There is no evidence that prizes crowd out adjacent technology areas; instead, they provide a modest crowd-in effect. Startups participating in an XPRIZE competition attract new investors who had not previously invested in the specific technology area addressed by the prize. Equity investors respond more than Public grant funders. Overall, the results suggest that participation in an innovation prize provides a positive signal to investors, especially in nascent technological areas where alternative signals aren't readily available.

JEL classifications: L26, G11, O31, M13

Keywords: Venture capital, Entrepreneurship, Investment decisions, Innovation prizes, New firms and startups

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†Saïd Business School, University of Oxford, UK. Emails: francois.bares.dphil@said.oxford.edu; thomas.hellmann@sbs.ox.ac.uk.

‡XPRIZE Foundation. Emails: elaine.hungenberg@xprize.org; andrew.tauhert@xprize.org.

1 Introduction

Science, technology, and innovation are central to economic growth (Aghion and Howitt, 1992; Romer, 1990). Yet the commercialisation of science, i.e., the progression from science to market, remains a significant economic and institutional challenge (Arora et al., 2020; Hall and Lerner, 2010). Funding plays a key role. At the science stage, funding mainly comes from research grants (Azoulay et al., 2019b; Howell, 2017). In the initial commercial stages, funding comes primarily from equity investors, especially venture capitalists (Hellmann and Puri, 2000; Kortum and Lerner, 2000), as well as angel investors, accelerators, and other seed-stage investors (Gonzalez-Uribe and Leatherbee, 2018; Hellmann, Schure and Vo, 2021; Kerr, Lerner and Schoar, 2014). Uncertainty is high at these initial stages. It limits investors' willingness to fund the first steps towards resolving the fundamental uncertainty of technologies that are potentially transformational (Ewens, Nanda and Rhodes-Kropf, 2018; Kaplan and Strömberg, 2004; Lerner and Nanda, 2020).

The primary economic mechanism for encouraging technology commercialisation is, naturally, the patent system. A large body of literature discusses the incentives and efficiency properties of patents (Arrow, 1962; Moser, 2005, 2013). The literature also recognises the potential value of patents in signalling quality and attracting investors (Conti, Thursby and Rothaermel, 2013; Farre-Mensa, Hegde and Ljungqvist, 2020; Hsu and Ziedonis, 2008, 2013). A fundamentally different approach to promoting innovation is via prizes. Prior literature (Kremer, 1998; Kremer and Williams, 2010; Wright, 1983) explores the incentive and efficiency properties of prizes relative to patents. A small empirical literature explores the design and economic effect of prizes, mostly finding that they increase innovation (Brunt, Lerner and Nicholas, 2012; Moser and Nicholas, 2013; Murray et al., 2012).

Our central research question in this paper is whether participation in an innovation prize helps startups attract new investments. This naturally focuses the analysis on the signalling role of prizes, and raises novel issues around their role in attracting investors.

We empirically investigate these questions using data from the XPRIZE Foundation. Over the period 1996 to 2023, it organised 25 prizes, including the Ansari XPRIZE for reusable spaceflight, XPRIZE Rapid Covid Testing, and XPRIZE Carbon Removal. We amalgamate data from XPRIZE and PitchBook to build a dataset capturing fundraising data for 416 startups participating in XPRIZE, as well as 4,607 startups in a control group. The latter are chosen as startups in PitchBook that share at least one of a prize's twenty leading participant keywords and recorded a PitchBook deal in the five years before launch. We follow financing through 2025Q2.

Our empirical estimation method leverages recent econometric advances in synthetic controls (Abadie, 2021). Specifically, we apply a Synthetic Difference-in-Differences (SDID) methodology, developed by Arkhangelsky et al. (2021) and previously used in economics and finance by Barwick et al. (2024), among others. The benefit of this method is that it optimises weights across all observations in the control sample to identify the most relevant benchmark composition that matches trends with the treatment sample during the pre-shock period. This allows for a targeted evaluation of any differential trends between treatment and control samples in the post-shock period. Our shock is the announcement date of the XPRIZE and the firms who successfully qualified to participate.

In our base model, we consider two main dependent variables, both of which capture equity fundraising by startups in our sample. The first is a quarterly indicator variable for whether a startup raised an equity round; the second is the cumulative count of equity rounds raised. We verify that treated and control observations have parallel pre-trends. The main finding is that XPRIZE participants (i.e., the treated group) have a significantly higher probability of raising equity. In the equity dummy specification, we find a treatment effect of 7.7%. Compared to a baseline of 4.7%, this suggests an increase of 7.7 percentage points (about 164% of that baseline) in the probability of raising equity in any one quarter. XPRIZE participants raise additional equity rounds as well. On average, participants raise 0.338 additional equity rounds, 23.4% more than the control group.

We then examine moderating effects to further understand what drives the main effects. We compare effects on the intensive margin, i.e., for startups that already raised equity before the prize announcement, versus the extensive margin, i.e., for startups that never raised equity beforehand. We conjecture that the effects should be largest on the extensive margin, because the prize's signalling value is higher for ventures with fewer credibility signals. Indeed, we find that the effects are stronger on the extensive margin. As noted above, patents already help startups signal their quality. We therefore conjecture that startups without patents show also benefit most from the XPRIZE signal. Dividing the sample into startups with and without patents, we again find that both subsamples continue to show significant equity fundraising effects, but, as conjectured, the effects are stronger for startups without patents. This suggests that XPRIZE participation is particularly impactful for startups lacking external signals regarding their potential. Crossing the two dimensions, participation raises the quarterly probability of raising equity by 12.2 percentage points for startups with neither prior equity nor a patent, against 3.0 points for startups with at least one of these signals. It is consistent with the notion that startups without any equity investments or patents are subject to particularly high uncertainty and therefore benefit most from the XPRIZE signal.

Next, we examine the importance of the prize purse, i.e., the headline value of how much money was pledged to the prize. We conjecture that larger prizes have a stronger signalling value. To test this, we divide the sample into larger and smaller prizes. Both subsamples continue to show significant equity fundraising effects, but the effects are stronger for larger prize purses. We also consider a simple measure of the 'expected prize', which divides the headline prize-purse by the number of participants (including those without a PitchBook match). We find that the effects are larger for larger expected prizes, though smaller prizes still show a positive and significant effect.

Our approach also recognises differences between patents and prizes, such as the fact that patents can be granted on an extremely wide range of discoveries that meet the generic patent requirements. Prizes, by contrast, are considerably narrower and can only be awarded to competition winners. If a prize thus draws attention to certain defined technological spaces, one might conjecture that it also detracts from adjacent spaces. We thus ask whether an XPRIZE competition takes the limelight away from such adjacent technologies. Our findings reject this conjecture. Specifically, we augment our control sample with a second control sample of technologically more distant startups. We find that all the main effects of the treated sample continue to hold when using the more distant control group. The difference between the adjacent and the more distant control groups is numerically small. Importantly, in the post-announcement period, the adjacent control sample exhibits slightly higher equity fundraising than the distant control sample, although this spillover is only about one-tenth of the participant effect on the probability of raising equity and one-fifth on the number of equity rounds. This suggests that, far from detracting, the XPrize seems to provide a small boost to equity fundraising in adjacent technology spaces.

Next, we ask whether the XPRIZE attracts not only new investments but also new investors. This changes our dependent variables, since we are no longer counting investment rounds, but distinct investors. If innovation prizes help alleviate the asymmetric information faced by investors, then the XPRIZE should help to attract the interest of previously disengaged investors. We consider three alternative definitions of what it means to engage new investors. First, we ask whether an investor is new to the startup, thus distinguishing between incumbent and new investors (Hellmann and Thiele, 2022). Second, we ask whether an investor is new to XPRIZE, i.e., whether they have ever invested in other startups that also participated in any XPRIZE competitions. Third, we inquire whether an investor is new to the prize's technological space. Across all three definitions, we find a consistent pattern: the effect of XPRIZE competitions is stronger for new investors than for incumbent investors. This provides a novel and nuanced understanding of the signalling

value of XPRIZE, suggesting that the signal it provides is more powerful among new investors who previously never invested in these types of startups and therefore may lack the insider knowledge that incumbent investors posses.

In addition to asking whether and how often startups raise equity, one may also ask how much money they raise. Unfortunately, the data is considerably weaker as funding amounts remain unreported for 45.0% of rounds in our sample. Notwithstanding this limitation, we find that our main model specification generates consistent results. That is, startups participating in XPRIZE competitions raise larger amounts of money.

Finally, we also ask whether prize participation also increases grant funding. We find this to be true, although the economic magnitudes are smaller than with equity fundraising. Public funders such as the US Department of Energy actively seek to fund nascent technologies, which carry the highest technological uncertainty, and should therefore have little need for an external signal (Azoulay et al., 2019a; Mazzucato and Semieniuk, 2017). Yet a body of evidence shows that public funders of science can be averse to uncertainty and favour less novel and ambitious projects (Azoulay and Greenblatt, 2025; Boudreau et al., 2016; Franzoni, Stephan and Veugelers, 2022). Our findings lie between these two views: public funders respond to the prize, but their response is about one-third to one-sixth of the equity response.

Needless to say, numerous robustness checks fortify our main empirical findings. In particular, the effects persist when we exclude prize winners and finalists, suggesting that they do not stem from prize money, and patenting in the prize technologies does not grow faster than in neighbouring technologies before launch, suggesting that XPRIZE does not simply pick technologies that already attract investors.

2 Institutional Setting and Data

2.1 XPRIZE competitions

The XPRIZE Foundation is a non-profit organisation that designs and manages large-scale public competitions intended to foster technological advancement and address significant societal or environmental challenges. The foundation operates on the premise that incentive competitions can accelerate innovation in sectors characterised by a high degree of technological uncertainty and capital intensity, often referred to as "deep tech".

The XPRIZE model gained global prominence with the launch of the Ansari XPRIZE in 1996, a $10 million competition challenging teams to build and launch a reusable spacecraft capable of carrying three people to 100 kilometres above the Earth's surface twice within

two weeks. Since then, the foundation has launched over 30 competitions with a total prize purse exceeding $500 million, targeting diverse sectors such as climate change, health, energy, and education.

Unlike traditional grant funding or venture capital, which provide funding to firms and projects based on their future potential, XPRIZE competitions reward demonstrated outcomes. A typical competition operates over a multi-year period, and follows a staged structure. At launch, the prize is announced with technical guidelines and success metrics, and teams are recruited globally, from university research groups and start-ups to established corporate teams. During qualification, registered teams submit technical documentation for a review of the viability of their approaches. During testing and elimination, teams are evaluated against objective performance metrics, and the field is narrowed to "Semifinalists" and then "Finalists". These stages often involve real-world prototype testing. For instance, the $100 million XPRIZE Carbon Removal competition evaluated how much carbon teams sequestered within a fixed time frame, subject to constraints on durability, scalability, and environmental impact. The competition culminates in the selection of winners who meet or exceed the final performance targets.

XPRIZE competitions provide a common review process with published criteria for technologies whose feasibility is difficult to assess. By establishing common performance criteria and evaluating different approaches against them, they create increasingly strong signals as teams participate and advance. The prizes are technology-agnostic: they reward outcomes regardless of the solution used. All tested outcomes are measurable, and a panel of independent experts assesses teams' performance against the announced criteria. Teams that pass the initial review become participants; some competitions subsequently narrow the field through semifinal and final stages before allocating milestone or final awards. Although the timing and technical requirements differ across prizes, launch, evaluation, and awards define the common sequence.

The focal event is participation at prize launch, rather than a later competition result. Launch reveals the problem, the assessment framework, and the set of participating firms. Participation entails more than simply expressing interest in a competition. Prospective teams submit information about their organisation, technical approach, and proposed pathway towards meeting the competition requirements. XPRIZE reviews these submissions for eligibility and conformity with the competition requirements before recognising teams as official competitors. The review is selective. Across the 25 competitions we study, 1,172 of 5,309 registered teams qualified (22.1%; Appendix Table A1). In XPRIZE Carbon Removal, 336 of 1,334 registered teams qualified. In IBM Watson AI XPRIZE, 68 of 458 did, and in the Next-Gen Mask Challenge, 28 of 991.

Qualification therefore sends two signals to investors. First, it certifies that the XPRIZE review found the firm's technical approach viable, a certification that fewer than one in four registered teams obtain. Second, it reveals the firm's willingness to have its technology measured by independent assessors against public criteria. A firm that doubts its technology has little reason to accept this exposure. The participation event thus identifies firms that have self-selected into a defined technological challenge, passed a selective review, and committed to external testing.

Participating in the prize is costly. To achieve the required technical milestones, participants often spend many multiples of the prize purse. The initial Ansari prize had a purse of $10 million, but participants spent over $100 million on R&D to meet the challenge. This suggests that the benefits of performing well exceed the expected monetary value of winning the prize. Other benefits include the signalling value to investors and other external stakeholders, media visibility, or the expected commercial gains from the technological breakthroughs that XPRIZE requires.

2.2 Data and sample construction

Our analysis needs two pieces of information for every firm: whether it qualified for an XPRIZE competition, and how it financed itself before and after the prize launch. XPRIZE records provide the first. They list each competition, its launch date and headline purse, and the teams that registered and were admitted as participants. The XPRIZE team linked each participant to its PitchBook company record by hand, tracing changes in firm names. PitchBook provides the second. It records each firm's founding date, headquarters, technology keywords, and patent counts, together with its financing rounds and the investors in each round. The technology keywords are tags that PitchBook adds to describe each firm's technology. We retain firms founded in or after 1990 and follow them until 2025Q2, the last quarter of the financing data.

We classify every deal with one mapping for participants and controls (Appendix A.3). Equity rounds comprise venture capital (VC) rounds, early non-VC rounds such as angel, seed, and accelerator rounds, and corporate equity. Grants form a separate class. We remove grants provided by XPRIZE, so that prize money never enters measured financing. We control for inflation and measure all monetary values in 2017 USD.

The choice of controls determines how we interpret our results. We want firms that investors could have financed instead of participants: firms working on similar technologies and active in financing markets, but absent from the competition. We therefore build a separate control pool for each prize. We take the twenty PitchBook keywords most

frequent among the prize's participants. A non-participant enters the pool if it shares at least one of these keywords and records a PitchBook deal[1] in the five years before prize launch. The keyword condition ensures that the control firms are near the prize's technology space. The deal condition confirms that the firm was active before launch, without conditioning on what happened afterwards. A firm eligible for several prizes joins only the pool with which its keywords overlap most (Appendix A.4). For robustness we also build an alternative control pool from patent classes. A non-participant enters it if it holds one of the twenty Cooperative Patent Classification (CPC) subclasses most frequent among the prize's participants and records a deal in the five years before launch.[2] Patent examiners assign CPC subclasses under a common scheme, so this pool is less arbitrary than one built from PitchBook keywords. It is, however, restricted to firms with patents, a selected group that includes only about half of participants (Table 1). We keep the keyword pool as the baseline. The CPC pool gives larger estimates (Section 4.2). Prizes with fewer than ten matched participants are bundled into an "Other" prize, in which each firm keeps its own prize's launch date, as this smaller prize does not have enough variation for an independent treatment effect estimate. To test whether prizes affect neighbouring technologies (Section 5.3), we also build a ring of distant firms. These firms share none of the participants' twenty leading keywords but share at least one keyword with the twenty most popular keywords among the control group, excluding those of the treatment group (Figure 1).

Appendix A.5 lists the leading keywords and contrasting control descriptions for the four largest participant cohorts.

The unit of observation is a firm, observed over the 25 quarters $\tau \in [-12, 12]$ around launch. A firm in two prizes would contribute two observations. In the estimation sample this occurs only once for participants, and it occurs for controls only inside the "Other" cohort.[3] For simplicity we therefore refer to these units as firms. Some participants first appear in PitchBook only after the prize launch, although XPRIZE records show that they existed at launch. We set their financing to zero before their first recorded deal and apply the same rule to controls.[4] The estimation sample contains 383 participant firms, 4,828 control firms, and 823 participant equity deals within the event window.

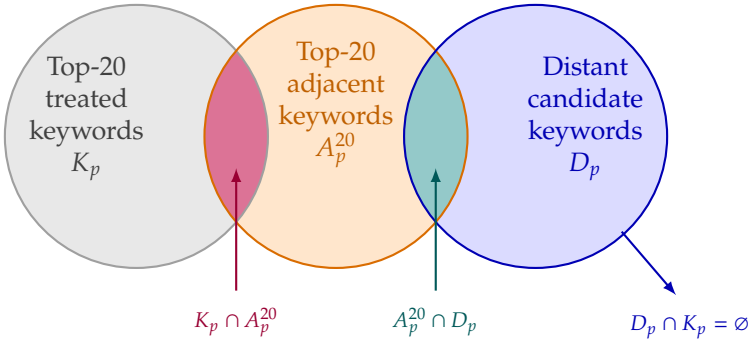
[1]The deal within five years could be a grant; it need not be an equity deal.

[2]Firm CPC subclasses come from PitchBook patent fields or, where these are missing, from matched USPTO patents.

[3]That cohort keeps each small prize's launch date. 214 control firms remain in more than one of these prizes, so the 4,828 control units are 4,607 firms. The 383 participant units are 383 firms.

[4]The rule never extrapolates beyond a recorded exit on PitchBook or beyond 2025Q2. Appendix A.2 describes the quarterly aggregation and the treatment of missing quarters.

Figure 1: Keyword sets for adjacent and distant controls



Notes: K_p is the set of the prize's twenty leading participant keywords, used to build the baseline control pool which we call adjacent. A_p^{20} is the set of the twenty keywords most frequent among these adjacent controls. D_p is the distant candidate pool's keyword set: firms that share a keyword in A_p^{20} but none in K_p, so that $D_p \cap K_p = \varnothing$ by construction. The distant ring is thus one step removed from the participants' own technology space.

2.3 Descriptive statistics

Table 1 shows that participants are young firms at the edge of equity markets. Of the 416 unique participant firms, just over half are headquartered in the United States (51.0%) and hold at least one patent (54.3%). Fewer than half had raised an equity round before launch (46.6%), and only 19.0% had raised a VC round (other equity rounds include angel, seed, and incubator rounds). In the estimation sample, 4.7% of participants raise equity in any given quarter. Most participants therefore enter a competition with a thin record of private financing, or none at all. This is the population for which an external assessment of technology should matter most. Their later trajectories remain open: 81.7% have no recorded IPO, acquisition, or bankruptcy by 2025Q2, with the caveat that exit data from PitchBook may be incomplete.

The investors that eventually finance participants are numerous and international. The 1,389 VC investors in participant firms are headquartered in 60 countries, and 56.0% of them are based in the United States. Their median assets under management are USD 142.5 million, and syndicates finance 39.7% of participant equity deals.

Prize participant are operating mostly in 'deep-tech', capital intensive sectors. The volume and type of grant support received by these firms attest to this characterization. Appendix Figure A1 classifies 531 of the participant deals as grants. The leading providers are predominantly public science and technology funders, including the National Science Foundation, the Department of Energy, NASA, the NIH and innovate UK (Appendix Table A7).

Figure 2 places the sample against the competition calendar. Participation accumulates slowly until 2014 and accelerates with the cluster of prizes from 2019 onwards. The

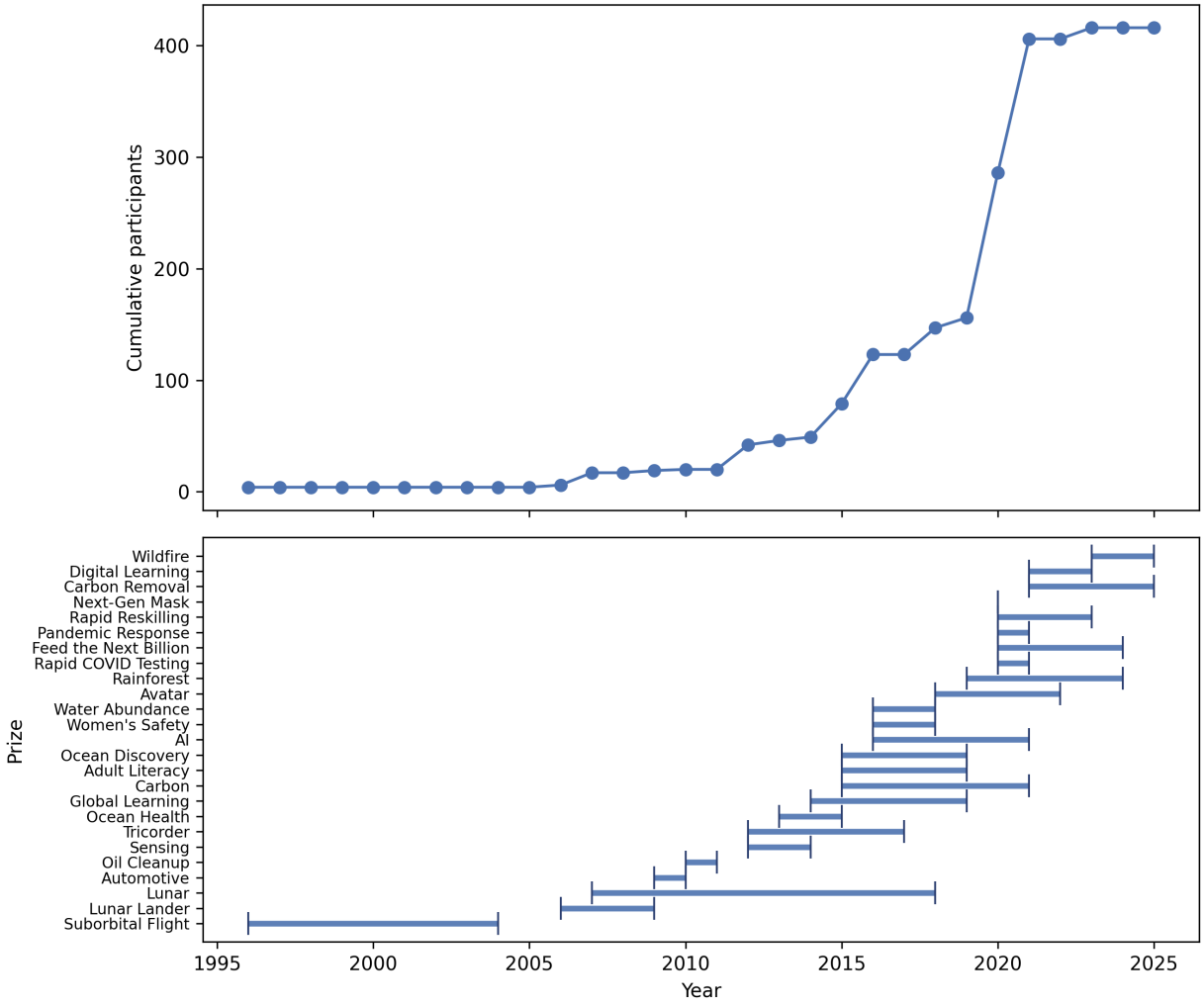
Table 1: XPRIZE participants and their VC investors

Participant firms	
Unique participant firms	416
US headquartered	51.0%
At least one patent	54.3%
Has pre-launch funding	51.7%
Probability of equity	4.7%
Has an equity round	46.6%
Has a VC round	19.0%
Has a grant round	20.9%
Has an accelerator/incubator round	24.0%
IPO	2.4%
Acquisition	8.7%
Failure	7.2%
No recorded exit	81.7%
VC investors in participant firms	
Unique VC investors	1,389
Investor countries	60
US headquartered	56.0%
Median AUM (real 2017 USD million)	142.5
Syndicated equity deals	39.7%

Notes: The participant rows use one observation per unique PitchBook-matched participant firm; the VC rows use one observation per distinct VC investor in participant firms. "At least one patent" indicates a positive patent count in the undated PitchBook snapshot. Funding, equity, VC, grant, and accelerator or incubator rows indicate a recorded round before launch. The "Probability of equity" row reports the mean equity dummy across participant units over the 12 pre-launch quarters of the estimation sample. Exit rows classify each firm once using records to 2025Q2, with precedence to IPO, then acquisition, then failure. AUM is assets under management in real 2017 USD. A deal is syndicated when it lists more than one investor.

competitions overlap and differ in length. The lounch of larger and more ambitious prizes in recent years makes our sample relatively loaded in the last few years of the recent sample window. As such we cannot have too long a time horizon post-prize in our estimation sample. We choose three years. Our estimates, therefore, are conservative and may not incorporate longer term effects on participant firms.

Figure 2: Participant coverage and XPRIZE competition timing



Notes: The upper panel reports the cumulative number of unique PitchBook-matched participant firms by year. Prize labels in the lower panel omit sponsor names and the XPRIZE/XCHALLENGE designation. The lower panel reports each competition's recorded period, ordered by launch, on the same horizontal axis. Vertical end marks delimit each period. Rows are spaced vertically for legibility.

3 Empirical Strategy

3.1 Stacked synthetic difference-in-differences

The central difficulty in estimating the effect of participation is selection. Firms choose whether to enter a competition, and XPRIZE chooses whom to admit. Participants may therefore differ from other firms in technology, quality, and financing history, both in levels and in trends. A firm that enters a prize may also be one that was about to raise capital anyway. A difference-in-differences comparison with the average non-participant would attribute such differences to the prize. Moreover, each prize is different and can be thought of as one independent shock in a specific technological space. There is no 'natural' control group that matches the trajectories and characteristics of each set of prize participants but does not itself participate. If they were so similar, they would have likely participated.

We address this problem by implementing a synthetic difference-in-differences (SDID) design (Arkhangelsky et al., 2021). Rather than assuming that participants and the average control would have moved in parallel, SDID constructs a control group that did move with participants before launch. It combines the unit reweighting of synthetic controls (Abadie, 2021) with the double differencing of difference-in-differences. Control units are reweighted so that their pre-launch outcomes track those of participants, and pre-launch quarters are reweighted so that they resemble the post-launch period. The effect of participation, the average treatment effect on the treated (ATT), is the divergence between participants and this synthetic control after launch.

Let i index firms and τ event quarters. Let C denote the N_0 control units, $\bar{Y}_{1\tau}$ the mean outcome of the N_1 participant units, and $T_0 = 12$ the number of pre-launch quarters. The control weights solve

$$(\widehat{\omega}_0, \widehat{\omega}) = \underset{\omega_0 \in \mathbb{R},\, \omega \in \Delta_{N_0}}{\arg\min} \sum_{\tau=-12}^{-1} \left(\omega_0 + \sum_{i \in C} \omega_i Y_{i\tau} - \bar{Y}_{1\tau} \right)^2 + \zeta^2 T_0 \| \omega \|_2^2, \tag{1}$$

where Δ_{N_0} is the unit simplex. The intercept ω_0 allows participants and their synthetic control to differ in levels; only their pre-launch paths must match. The ridge penalty spreads weight across many controls, so that no small group of firms drives the comparison. Time weights $\widehat{\lambda}_\tau$ solve the analogous problem across pre-launch quarters, so that the weighted pre-launch outcomes of controls reproduce their post-launch mean:

$$\left(\widehat{\lambda}_0, \widehat{\lambda}\right) = \underset{\lambda_0 \in \mathbb{R},\, \lambda \in \Delta_{T_0}}{\arg\min} \sum_{i \in C_p} \left(\lambda_0 + \sum_{\tau=-12}^{-1} \lambda_\tau Y_{ip\tau} - \bar{Y}_{ip,\text{post}} \right)^2 + \zeta_\lambda^2 N_{0p} \| \lambda \|_2^2, \tag{2}$$

Following the typical double differencing of the traditional DID estimator, the treatment effect is then estimated as:

$$\widehat{\tau}^{\text{sdid}} = \left(\bar{Y}_{1,\text{post}} - \sum_{\tau=-12}^{-1} \widehat{\lambda}_\tau \bar{Y}_{1\tau}\right) - \sum_{i \in C} \widehat{\omega}_i \left(\bar{Y}_{i,\text{post}} - \sum_{\tau=-12}^{-1} \widehat{\lambda}_\tau Y_{i\tau}\right), \tag{3}$$

where $\bar{Y}_{1,\text{post}}$ and $\bar{Y}_{i,\text{post}}$ are means over the 13 quarters $\tau \in [0, 12]$. The first term measures how much participants' outcomes change from the weighted pre-launch period to the post-launch period. The second term measures the same change for the synthetic control. Their difference is the average treatment effect on the treated (ATT), denoted $\widehat{\tau}^{\text{sdid}}$. For more details on how the estimator is derived and how its bias vanishes when either set of weights reproduces its target, see Arkhangelsky et al. (2021).

Prizes launch in different years, so we estimate Equation (3) on a stacked panel. Every cohort is aligned on its own launch quarter, and every firm observation receives its own identifier[5]. Treatment equals one for participants from $\tau = 0$ onwards[6]. The stacked estimator yields one treatment effect per outcome across all prizes, but we also estimate the effect for each prize separately in Appendix C.6.

3.2 Outcomes and inference

We measure equity fundraising in two ways, which capture different margins. The *equity dummy* equals one if the firm raises at least one equity round in quarter τ. Its treatment coefficient is the change in the quarterly probability of raising equity, in percentage points. It tells us whether participation changes how often firms raise equity, if at all. The *equity deal count* is the cumulative number of equity rounds raised since $\tau = -12$, in levels. Its treatment coefficient is the number of additional rounds per participant, averaged over the post-launch quarters, relative to the weighted control group, and tells us whether these additional rounds accumulate. The count complements the dummy variable in two ways. First, it retains multiple rounds raised in a quarter, which the dummy records as a single financing quarter. Second, averaging cumulative counts also makes the estimate sensitive to timing: an additional round at launch enters all 13 post-launch quarters, whereas one in the final quarter enters only one. Sections 6 and 7 introduce the investor and grant outcomes.

Standard errors come from 50 unit-bootstrap replications. Each replication resamples

[5] It is extremely rare for the same firm to participate in more than one XPRIZE, but when it happens, the firm is effectively treated as a different firm in each prize.

[6] Estimation uses version 0.10.1 of the Python implementation `synthdid` (D2CML Team et al., n.d.).

complete 25-quarter firm histories with replacement and re-estimates both sets of weights. Each replication reweighs the control group and time weights with the new sample. Many of our tests compare the treatment effect across different subsamples A and B of the treatment group, for example, participants with and without a patent. The two estimates share the same control group (though with different weights) and are therefore correlated, so comparing their separate confidence intervals would mislead. We instead re-estimate both treatment effects on the same resampled units in each replication and record $\widehat{\Delta}^{(b)} = \widehat{\tau}_A^{(b)} - \widehat{\tau}_B^{(b)}$.[7] The standard error of $\widehat{\Delta} = \widehat{\tau}_A - \widehat{\tau}_B$ is the standard deviation of the 50 paired differences.

3.3 Identification

The key assumption is that, without participation, participants would have followed the post-launch path of their synthetic control. This assumption cannot be tested after launch, but its pre-launch counterpart can: the weighted paths should coincide before launch. Figure 3 in Section 4.1 shows that they do. Beyond the validity of the counterfactual, three key challenges to our interpretation nonetheless deserve attention.

The first challenge is anticipation. XPRIZE might target technological spaces that are on the ascendancy and are increasingly attracting investors and talent. The second challenge is market spillovers. Control firms operate in technology spaces that overlap with those of participants. This means they may share the same pool of potential investors. If a prize diverts investors away from, or towards, these firms, the control path is contaminated. Section 5.3 tests this directly. The third challenge is that participation bundles several treatments: screening, public exposure, technical validation and, crucially, a chance of winning prize money. Winning a large monetary prize for a firm should, in principle, reduce its cost of capital and increase both its ability and willingness to raise equity, irrespective of the certification effect we try to identify. Section 5.2 addresses this point. We address all these concerns in later sections.

Measurement also limits interpretation. PitchBook coverage of early-stage deals may be incomplete, and we treat a quarter with no recorded PitchBook deal as a quarter with no financing. We have no reason to expect this undercoverage to differ systematically between participants and controls, although it may reduce precision. Funding amounts are a larger problem. PitchBook reports no amount for about 45% of participant equity deals in the

[7] Each replication draws firms with replacement from the pooled sample, participants in both subsamples and controls together, and keeps each unit's 25-quarter path. It then splits the draw into subsamples A and B. Controls are resampled in that draw and retained in both subsamples; each subsample re-estimates its own weights. A draw that leaves either subsample without a treated or a control unit is discarded.

event window, and the missing amounts are not random across firms. We therefore focus on fundraising incidence and deal counts. We still estimate the effect on amounts raised and on valuations, and report these estimates in Appendix D. Participation raises total recorded equity capital by 9.7%, significant at the 1% level, and the first recorded equity round by 11.3%, significant at the 5% level. The estimated post-money valuation effects are 42.5% for participants with prior equity rounds, significant at the 5% level, and 82.9% for participants without, significant only at the 10% level. These estimates carry the caveats of the underlying samples. The amount measures either set unreported amounts to zero or drop units, and PitchBook records valuations for only 58 and 16 participant units. We therefore treat them as supporting evidence and do not discuss them further in the main text.

4 Prize Participation and Equity Fundraising

We begin with the central question of the paper: do start-ups raise more equity after they enter an XPRIZE competition? We first estimate the average effect of prize participation for all firms, then test its robustness, and finally ask which firms benefit most.

4.1 Baseline effect

Table 2 answers the central question in the affirmative. After launch, participation raises the quarterly probability of raising equity by 7.7 percentage points. The magnitude is large relative to where participants start. Before launch, they raise equity in only 4.7% of quarters (Table 1), so the effect is 1.64 times their pre-launch incidence. The deal count confirms the result in cumulative form. Over the three post-launch years, participants completed 1.781 equity rounds on average, against 1.443 for their synthetic control. The difference of 0.338 rounds amounts to 23.4% more equity rounds. Both estimates are significant at the 1% level.

Figure 3 shows how the effect unfolds and supports the design behind it. Before launch, the participant and synthetic-control paths are indistinguishable for both outcomes. They move together even through the rise in fundraising in the two quarters before launch. The paths separate at launch. Participants continue to raise equity in 8% to 14% of quarters, whereas the incidence of their synthetic control falls from 6% to about 1%. Cumulative rounds diverge from the second post-launch quarter, and the gap widens until the end of the window.

The persistence of this gap shapes its interpretation. A burst of publicity at launch

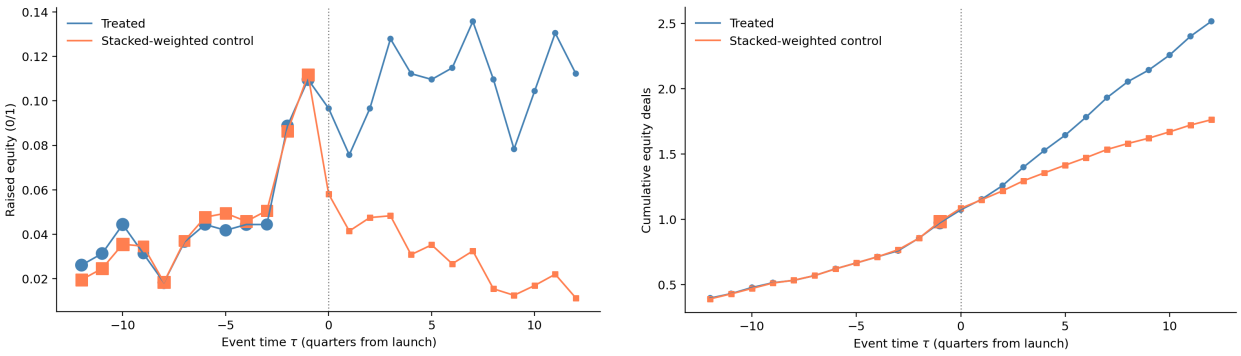
Table 2: The effect of XPRIZE participation on equity fundraising

Estimate	Equity dummy	Equity deal count
Baseline sample		
ATT	0.077^{***}	0.338^{***}
	(0.006)	(0.047)
N treated firms	383	383
N treated deals	823	823

Notes: Entries are event-time stacked synthetic difference-in-differences estimates of the treatment effect on the treated (Equation (3)). Parentheses contain standard errors from 50 unit-bootstrap replications that resample complete firm histories. N treated firms are participant firms. N treated deals are participant equity deals within the event window $\tau \in [-12, 12]$. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The equity dummy equals one if the firm raises at least one equity round (VC, early non-VC, or corporate equity) in the quarter; its treatment effect is a change in the quarterly probability of raising equity, so that 0.077 corresponds to 7.7 percentage points. The equity deal count is the cumulative number of equity rounds since $\tau = -12$, in levels; its treatment effect is the number of additional rounds per participant unit, averaged over the post-launch quarters. The control group contains 4,828 firms, drawn from non-participants that share at least one of the prize's twenty leading participant keywords and record a deal in the five years before launch.

Figure 3: Equity fundraising of participants and their synthetic control

(A) Equity dummy (B) Equity deal count



Notes: The figure plots, for each event quarter τ relative to prize launch, the mean outcome of the 383 participant firms (blue) and of their SDID-weighted synthetic control drawn from 4,828 adjacent control units (orange). Panel 3(A) reports the equity dummy, which equals one if the firm raises at least one equity round in the quarter. Panel 3(B) reports the cumulative number of equity rounds since $\tau = -12$. The synthetic control applies the unit weights of Equation (1), including the fitted intercept. Marker size before launch is proportional to the SDID time weight of the quarter. The vertical line marks the launch quarter $\tau = 0$.

would raise fundraising briefly and then fade. Instead, participants remain on a steeper financing path for three years. This pattern fits better with a lasting change in what investors know about participants and their technologies. The remainder of the paper examines this interpretation.

4.2 Robustness

We further qualify the baseline estimates by showing that they are robust to (i) alternative control group constructions; (ii) pre-trend fundraising among the treated; (iii) the type of equity investors; and (iv) outlier prizes that might drive the effect.

The first check concerns the control pool. The baseline selects controls with PitchBook keywords, which PitchBook assigns from firm descriptions by machine learning and manual annotation. These classifications may be arbitrary. An alternative pool relies on patent classes instead. It contains non-participants that hold at least one of the twenty CPC subclasses most common among the prize's participants (Section 2.2). Against this pool, participation raises the quarterly probability of raising equity by 8.8 percentage points and adds 0.516 rounds, both significant at the 1% level (Appendix Table A8). Within the baseline pool, the estimate does not rest on a few heavily weighted controls. For the equity dummy, all 4,828 control units receive positive weight, and the effective number of controls is 4,678 (Appendix Figure A7).

The second check concerns the rise in fundraising before launch. Participants raise equity in 4.4% of quarters at $\tau = -3$, 8.9% at $\tau = -2$, and 11.0% at $\tau = -1$ (Panel 3(A) of Figure 3). The synthetic control reproduces this spike, but the spike raises a concern. Firms may enter a prize because they are about to raise capital, or because their technology already attracts investors. The estimated effect would then partly reflect plans made before launch. The estimates do not rest on these firms. We randomly remove 70% of the participant units with exactly one equity round in $\tau \in \{-2, -1\}$ and re-estimate the SDID weights. The spike largely disappears. The remaining participants raise equity in 4.4% of quarters at $\tau = -3$, 5.8% at $\tau = -2$, and 4.4% at $\tau = -1$ (Appendix Figure A10). The effects are 8.8 percentage points and 0.349 rounds, both significant at the 1% level (Appendix Table A10). Removing half of these units gives 8.8 points and 0.353 rounds (Appendix Table A9). Section 5.4 examines what drives the spike.

The third check concerns the type of investor. Equity rounds include rounds from angels, accelerators, and incubators, not only from VCs. VC rounds account for about half of participant equity rounds in the event window: 425 of 823. The results, however, do not rest only on non-VC rounds. Counting only VC rounds as equity, participation raises

the quarterly probability of raising equity by 6.5 percentage points and adds 0.165 rounds, both significant at the 1% level (Appendix Table A17).[8]

The fourth check concerns heterogeneity across prizes. Estimated prize by prize, fundraising incidence is positive and significant in all eight cohorts, but its size varies greatly in magnitude (Appendix Table A15). It ranges from 3.4 percentage points in XPRIZE Rapid Covid Testing to 10.5 points in XPRIZE Carbon Removal. The deal-count effect is significant in only three prizes: IBM Watson AI XPRIZE, XPRIZE Carbon Removal, and XPRIZE Feed the Next Billion. XPRIZE Carbon Removal, the largest prize, combines the largest incidence effect with the largest weight in the estimates, as it contributes 32% of the treatment observations. However, the treatment effects without the Carbon Removal prize remain large and significant. Excluding its Carbon Removal participants and controls yields a treatment effect of 6.2 percentage points and 0.256 rounds, both significant at the 1% level (Appendix Table A14). Section 5.2 further discusses prize heterogeneity in terms of prize size.

4.3 Extensive and intensive margins

To better understand how the prize affects financing, we distinguish first-time equity fundraisers from firms with prior equity rounds. We split participants into the 222 units without an equity round before launch, the extensive margin, and the 161 units with at least one, the intensive margin.

Table 3 presents the results. On the extensive margin, participation raises the quarterly probability of raising equity by 11.9 percentage points, against 2.4 points on the intensive margin. The effects are significant at the 1% and 5% levels, respectively. The intensive-minus-extensive difference is -9.4 points and significant at the 1% level. The deal count rises by 0.375 rounds on the extensive margin and 0.289 on the intensive margin, both significant at the 1% level. The difference in deal counts is not statistically significant.

The incidence effect is larger for firms without prior equity, consistent with participation easing first-time fundraising. Both groups raise more rounds, but their deal-count effects cannot be distinguished statistically. These contrasts compare effects relative to separately fitted synthetic controls; they do not by themselves identify why the incidence effects differ. Section 5.1 interprets this difference in terms of signals.

[8] A VC round has at least one VC investor. The VC-only sample retains the 202 participant units and 2,579 control units with at least one VC round in the event window.

Table 3: Equity fundraising on the intensive and extensive margins

Estimate	Equity dummy	Equity deal count
intensive margin		
ATT	0.024**	0.289***
	(0.010)	(0.063)
N treated firms	161	161
N treated deals	501	501
extensive margin		
ATT	0.119***	0.375***
	(0.008)	(0.063)
N treated firms	222	222
N treated deals	322	322
Intensive − Extensive		
Difference	-0.094***	-0.086
	(0.010)	(0.094)

Notes: The intensive margin contains participant units with at least one equity round before prize launch. The extensive margin contains participant units with none. The two groups partition the baseline participant units. Both estimates use the same 4,828 eligible control units, with SDID weights fitted separately for each margin. The difference row reports the intensive-margin effect less the extensive-margin effect, with standard errors from 50 paired bootstrap draws that resample the same firm–prize units in both margins.

5 Why Does Participation Matter? Prizes as Signals

The results so far establish that participation in an XPRIZE helps early-stage startups to raise funds. They do not establish why. Our hypothesis is that a prize produces information. It announces a technical problem, screens entrants against publicly announced criteria, and places admitted firms under expert and, crucially, independent scrutiny. Technological uncertainty limits the willingness of investors to fund early-stage firms (Ewens, Nanda and Rhodes-Kropf, 2018), and this information reduces that uncertainty. The prizes can produce a powerful certification effect and alleviate technological uncertainty as perceived by investors. If the interpretation is correct, two further patterns should appear. First, the effect should shrink for firms that already hold other credible signals, such as prior equity rounds or patents. The larger effect on the extensive margin (Section 4.3) gives a first indication. Second, the effect should grow with the stakes of the competition, because a more valuable prize attracts more scrutiny and stronger rivals. We examine each prediction in turn. We then address an alternative explanation to our results: XPRIZE may be picking technologies that were already attracting investors, and we ask whether the effect merely reflects this choice.

5.1 Prior equity, patents, and the prize signal

The first prediction concerns substitutes. If the prize informs investors about a firm and its technology, it should matter less for firms that already have credible signals about the quality of their technology. A prior equity round is one such signal: an investor has already screened the firm and committed capital to it. Section 4.3 is consistent with this prediction. Participation raises the quarterly probability of raising equity by 11.9 percentage points for firms without prior equity, against 2.4 points for firms with it. Firms with and without prior equity may, however, differ in other respects, so this contrast alone does not identify a signal. Patents provide a second test. A granted patent certifies novelty and credibility through an external examination and can act as a powerful signal to investors (Conti, Thursby and Thursby, 2013; Hsu and Ziedonis, 2013). If prizes and patents both reduce technological uncertainty for investors, participation should add less for firms that held a patent before the prize. We therefore compare participants with and without a pre-launch granted patent. We then cross patent ownership with prior equity.

Table 4 shows a larger incidence effect among units without an observed pre-launch patent. Participation raises the quarterly probability of raising equity by 8.6 percentage points for these firms, against 4.0 points for thoses with a patent before the prize.

The deal count has the same ordering: units without an observed pre-launch grant

Table 4: Equity fundraising by pre-launch patent ownership

Estimate	Equity dummy	Equity deal count
No patent		
ATT	0.086^{***}	0.372^{***}
	(0.008)	(0.048)
N treated firms	315	315
N treated deals	655	655
Patent		
ATT	0.040^{***}	0.180^{**}
	(0.010)	(0.071)
N treated firms	68	68
N treated deals	168	168
No patent − Patent		
Difference	0.045^{***}	0.192^{**}
	(0.013)	(0.083)

Notes: Entries are stacked SDID estimates of the treatment effect, with standard errors from 50 unit-bootstrap replications in parentheses. Outcomes and support rows follow Table 2. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. "Patent" contains participant units with at least one recorded patent granted strictly before the prize launch date. "No patent" means no pre-launch grant is observed. The extract covers 186 of the 383 participant units; 68 have an observed pre-launch grant. The difference row reports the no-patent effect less the patent effect, with paired-bootstrap standard errors. Controls are unchanged across groups.

add 0.372 rounds, against 0.180 for units with one. The difference of 0.192 rounds is significant at the 5% level. For both outcomes the effects for firms without a patent are approximately twice as large as for firms with one, suggesting that the prize acts as a certification mechanism that can substitute for some of the signal that patents provide.

Table 5 crosses patent ownership with pre-launch equity fundraising. It compares firms with neither signal with a pooled group holding at least one pre-launch signal, then reports the four cells and their pairwise contrasts.

Firms with neither signal respond most. Participation raises their quarterly probability of raising equity by 12.2 percentage points, against 3.0 points for firms with at least one signal. The two signals do not carry the same weight. Firms with a patent but no prior equity gain 8.4 points, against 2.5 points for firms with prior equity but no patent. Among firms without prior equity, the incidence effect is 3.8 points lower for patent holders, significant at the 5% level, although only 18 of these firms hold a patent. Among firms with prior equity, the patent difference is 0.1 points and not statistically significant.

These results qualify the patent comparison of Table 4. They suggest that that, although both patents and prior equity serve as signals to investors, their effects are not perfect substitutes. The incidence effect is lower with prior equity than with a patent, suggesting that a financing record substitutes for the prize signal more than a patent. Participation adds most for firms that enter with neither, consistent with the prize providing information that investors cannot obtain from other signals.

5.2 Prize purse and expected prize

The second prediction concerns the stakes. A larger purse draws more attention and stronger competitors, and a place in such a competition says more about a firm. Panel A of Table 6 splits participants at a total prize purse of USD 10 million. Participation raises fundraising in both groups, but the effect is markedly stronger in large prizes. The incidence effect is 9.4 percentage points in large prizes against 5.6 in small ones, a difference of 3.8 points that is significant at the 1% level. The contrast in the deal count is sharper still: 0.466 additional rounds against 0.192, more than twice as many.

Total purse, however, is an imperfect measure of what participants can expect to win. Larger prizes also attract more teams. Across 25 prizes, each additional USD 1 million of purse is associated with 3.1 more participating teams (Appendix Figure A3). A large purse shared among many rivals may promise each of them less than a modest purse contested by few. Panel B therefore replaces the headline purse with the expected prize, the purse divided by the number of participating teams. We split prizes at USD 0.200 million per

Table 5: Equity fundraising by pre-launch equity and patent status

	Raised equity (0/1)	Equity deal count
Prior equity, patent	0.026*	0.203**
	(0.016)	(0.099)
N treated units	50	50
Prior equity, no patent	0.025**	0.326***
	(0.010)	(0.093)
N treated units	111	111
No prior equity, patent	0.084***	0.142
	(0.020)	(0.152)
N treated units	18	18
No prior equity, no patent	0.122***	0.396***
	(0.007)	(0.065)
N treated units	204	204
At least one signal	0.030***	0.270***
	(0.007)	(0.060)
N treated units	179	179
Neither − at least one signal	0.092***	0.126
	(0.010)	(0.086)
Prior equity: patent − no patent	0.001	-0.123
	(0.020)	(0.150)
No prior equity: patent − no patent	-0.038**	-0.254
	(0.019)	(0.159)
Patent: prior equity − none	-0.058**	0.061
	(0.024)	(0.162)
No patent: prior equity − none	-0.098***	-0.070
	(0.013)	(0.116)
No prior equity, patent − prior equity, no patent	0.059***	-0.184
	(0.023)	(0.187)

Notes: Entries are stacked SDID effects on the quarterly probability of raising equity and the cumulative number of equity rounds, with standard errors from 50 paired firm-unit bootstrap replications in parentheses. Prior equity means at least one equity round before prize launch; no prior equity means none. Patent means at least one observed patent granted strictly before launch; no patent means no pre-launch grant is observed. "Neither" means neither signal is observed. "At least one signal" pools the other three participant cells and is fitted as one treated group, not an average of their effects. Every group uses the same 4,828 control firm–prize units, regardless of their equity or patent status; SDID weights are fitted separately for each group. The difference row reports neither less the pooled group using paired bootstrap draws. N treated units counts participant firm–prize units. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 6: Equity fundraising by total and expected prize purse

Panel A: Total prize purse

Estimate	Equity dummy	Equity deal count
Small purse (< $10 million)		
ATT	0.056***	0.192***
	(0.007)	(0.045)
N treated firms	194	194
N treated deals	342	342
Large purse (≥ $10 million)		
ATT	0.094***	0.466***
	(0.010)	(0.075)
N treated firms	189	189
N treated deals	481	481
Large - Small		
ATT	0.038***	0.274***
	(0.010)	(0.067)

Panel B: Expected prize purse

Estimate	Equity dummy	Equity deal count
Small purse (≤ $0.200 million)		
ATT	0.058***	0.189***
	(0.009)	(0.058)
N treated firms	175	175
N treated deals	304	304
Large purse (> $0.200 million)		
ATT	0.091***	0.455***
	(0.009)	(0.063)
N treated firms	208	208
N treated deals	519	519
Large - Small		
ATT	0.033***	0.267***
	(0.012)	(0.085)

Notes: Entries are stacked SDID estimates of the treatment effect, with standard errors from 50 unit-bootstrap replications in parentheses. Outcomes and support rows follow Table 2. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Panel A classifies total prize purses in real 2017 USD as small below USD 10 million and large at or above that threshold. Panel B classifies the expected prize purse per participating team, calculated by dividing the total prize purse by the number of unique, non-excluded teams in the XPRIZE team roster, as small at or below USD 0.200 million and large above that threshold. Each "Large - Small" row reports the effect for the large-prize group minus that for the small-prize group; standard errors use paired bootstrap draws that re-estimate both effects on the same resampled units. Controls are unchanged across groups.

team, which divides participating teams almost equally, with 629 below and 651 above (Appendix Tables A5 and A6). The estimates are almost unchanged from the total purse split. Above the threshold, incidence rises by 9.1 percentage points and the deal count by 0.455 rounds. At or below it, the effects are 5.8 points and 0.189 rounds. Both differences are significant at the 1% level.

The two panels thus tell the same story. Investors respond more to participation in competitions that are more valuable, whether value is measured in total or per competitor. However, the gap between small and large prizes can also be interpreted differently: investors may respond to prize money rather than to information. The larger the prize, the more cash the firm has to invest that does not have to come from further diluting its equity. This should increase the ability and willingness of the firm to raise additional funding. While winners receive the largest prizes and finalists also typically receive a prize, even if they do not, their expected prize winnings conditional on making the final are much larger than the unconditional prize purse. If money drove the effect, removing these firms should weaken it. It does not. Excluding the 15 winner units leaves a treatment effect of 7.8 percentage points and 0.344 rounds. Excluding the 74 winners and finalists leaves 8.3 points and 0.420 rounds (Appendix Table A11). The effect arises at participation and does not come from the firms that end up with prize money. Furthermore, Across 18 prizes, the ratio of recorded equity and grant funding to the total purse averages 8.22 in the twelve quarters before launch and 15.75 in the twelve quarters from launch (see Appendix Table A13). Such a large discrpancy between fundraising and prize money suggests that the manetary value of the prize is not the chief motivation for firms to participate and win.

The stronger effects among larger prizes raise a question: Is there a critical prize size above which participation improves fundraising significantly? Or is there a 'floor' treatment effect that applies to all prizes? With a traditional difference-in-differences, one could interact the treatment with prize size to answer this question. SDID has no such term. What we can do instead is estimate the SDID effect for each prize and then modify the weighting scheme described in Arkhangelsky et al. (2021) to include weights proportional to total purse and expected purse. More specifically, the standard procedure weights each prize by its number of participants. We keep this weight, but add to it a weight that is inversely proportional to total purse size. Then we add a weight that is proportional to expected prize size. The details of the procedure can be found in Appendix B.1.

Table 7 shows that downweighting large prizes lowers the effect but does not remove it. Under standard weighting, the effects are 7.3 percentage points and 0.327 rounds.[9] Inverse-

[9]The standard cohort-weighted estimate differs from the stacked estimate of Table 2. It fits separate weights within each prize and retains only 18 raw prize cohorts.

Table 7: Equity fundraising under prize-size standardisation

Estimate	Equity dummy	Equity deal count
Standard cohort weighting		
ATT	0.073***	0.327***
	(0.008)	(0.051)
Purse prize weighting		
ATT	0.048***	0.174**
	(0.008)	(0.072)
Expected prize weighting		
ATT	0.044***	0.148**
	(0.008)	(0.069)
N prize	18	18
N treated firms	364	364
N treated deals	817	817

Notes: Entries aggregate prize-level SDID estimates $\widehat{\theta}_p$ with the normalised weight $w_p/\sum_q w_q$ of Equation (6). Standard cohort weighting sets $w_p = N_{1p}T_{1p}$. Purse prize weighting sets $w_p = N_{1p}T_{1p}/P_p$. Expected prize weighting sets $w_p = N_{1p}T_{1p}/(P_p/R_p)$. N_{1p} is the number of participant units in prize p, T_{1p} its number of post-launch quarters, P_p its headline purse in real 2017 USD, and R_p its number of participating teams. P_p/R_p is the expected prize. Appendix Table A12 reports the weight of each prize under each scheme. Parentheses contain standard errors from 50 unit-bootstrap replications. N counts the raw prizes with separately estimable cohorts. The equity dummy and equity deal count follow Table 2. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Note that the effect with the regular weighting differs from the baseline effect shown in Table 2 because we estimate the effect with a cohort weighting procedure rather than stacking the data from all prizes together.

purse weighting gives 4.8 points and 0.174 rounds. Inverse-expected-prize weighting, which assigns 51.5% of the weight to XPRIZE Rapid Covid Testing, gives 4.4 points and 0.148 rounds. The incidence estimates are significant at the 1% level and the count estimates at the 5% level. Assuming that the effects are linear, we can extrapolate from these estimates that a prize with zero purse would have a participation effect of 4.2 percentage points and 0.9 points per additional USD 10 million. See more details in Appendix B.1.

The evidence therefore separates two components. Participation raises equity fundraising whatever the size, value, or prestige of the prize. The effect survives the exclusion of award recipients, the exclusion of the largest prize, and a weighting that evaluates it at the purse of a small prize. Large, high-profile prizes amplify this effect. We attribute this to the greater signalling value of larger prizes, rather than their greater monetary value, as removing the winners and finalists from the sample does not weaken the participation effect.

5.3 Technological spillovers: does the prize affect adjacent firms?

Although prizes may substitute for some of the signalling value of a patent, they may operate at different levels. A patent signals the value of a particular invention and potentially, by extension, a particular firm. A prize might signal the value of a firm in a similar way, but it might also operate at a higher level. It might signal the promise of a technological space more broadly. This focus raises a concern. If investors move towards the technologies a prize highlights, they may move away from neighbouring ones, such that firms in the adjacent technological space are adversely affected. Such crowding out would matter twice. It would impose a hidden cost on nearby technologies. It would also inflate our estimates, because adjacent firms form our control group: a fall in their fundraising would widen the gap with participants even if participants gained nothing. In econometric terms, this would violate the stable unit treatment value assumption (SUTVA). Alternatively, the prize could generate positive spillovers for adjacent technologies by highlighting the promise of the broader technological space.

Table 8 examines this concern in two steps. First, we replace adjacent controls with distant controls, which are two steps removed from the participants' technological space (see figure 1). If crowding out inflated the baseline, the effect should shrink against these firms. It does not. The effects are 8.3 percentage points and 0.389 rounds, slightly above the baseline values of 7.7 points and 0.338 rounds. Second, we treat adjacent firms (those making up the control group in the baseline results), as if they were the treatment and compare them directly with distant firms. After launch, adjacent firms raise equity 0.7

percentage points more often and add 0.061 rounds than distant firms. Both estimates are significant at the 1% level.

Table 8: Equity fundraising and technological distance

Estimate	Equity dummy	Equity deal count
Treated vs adjacent controls		
ATT	0.077***	0.338***
	(0.006)	(0.047)
N treated firms	383	383
N treated deals	823	823
Treated vs distant controls		
ATT	0.083***	0.389***
	(0.005)	(0.051)
N treated firms	383	383
N treated deals	823	823
Adjacent vs distant controls		
ATT	0.007***	0.061***
	(0.002)	(0.017)
N treated firms	4,828	4,828
N treated deals	9,740	9,740

Notes: Entries are stacked SDID estimates of the treatment effect, with standard errors from 50 unit-bootstrap replications in parentheses. Outcomes and support rows follow Table 2. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Adjacent controls share at least one of the twenty most common keywords among participants. Distant controls share none of these keywords but share at least one of the twenty most common keywords of the prize's adjacent controls. The first two panels compare participants with adjacent and distant controls. The third panel assigns the prize launch as a pseudo-treatment to the adjacent firms and compares them with distant firms; its support rows count adjacent firms and their equity deals.

The prize therefore does not generate a crowd-out of investment from adjacent technologies. If anything, it provides some positive spillover. The spillover is small, about one-tenth of the participant effect on incidence and one-fifth on the deal count, but is significant at the one percent level. It suggests that part of what the prize conveys concerns the promise of a technology as a whole, and not only the quality of the admitted firms. It also implies that our baseline estimates, which use adjacent firms as controls, are if anything conservative.

5.4 Technological momentum: does XPRIZE pick hot technologies?

Panel 3(A) of Figure 3 shows that fundraising accelerates before launch. Participants raise equity in 2.6% of quarters at $\tau = -12$, 4.4% at $\tau = -3$, and 11.0% at $\tau = -1$. Section 4.2

shows that the estimates are robust to this pre-trend. Removing 70% of the participant units with one equity round in $\tau \in \{-2, -1\}$ flattens it and leaves a treatment effect of 8.8 percentage points and 0.349 rounds. The rise nonetheless calls for an explanation. Two mechanisms can produce it. They have different implications for our estimates.

The first mechanism concerns the firm. A firm that has just closed a round has cash and runway. It can commit resources to a demanding competition that may require substantial R&D investment, without raising capital at the same time. Such firms are more likely to enter. The design absorbs this selection. The synthetic control reproduces the pre-launch rise, so it consists of firms that also raised capital just before launch. After a round, firms with runway raise less, and the synthetic control does so: its incidence falls from 11.2% at $\tau = -1$ to 5.8% at $\tau = 0$. Participants do not follow this path.

The second mechanism concerns the technology. XPRIZE may launch competitions in technology spaces that already attract growing investor interest. Participants would then raise more capital after launch regardless of the prize. Adjacent controls operate in related spaces and may share such trend. The estimates are biased only if the boom concentrates narrowly in the participants' own technology.

The timing of prize design makes the deliberate choice of 'riding trends' unlikely. Once XPRIZE selects the topic of a competition, it can take months to secure millions of dollars from sponsors, draft its rules and technical guidelines, and organise the competition before the prize officially begins. The fundraising spike, by contrast, is confined to the two quarters before launch (Panel 3(A) of Figure 3). When XPRIZE chose the topic of most competitions, this spike had not yet occurred and could not guide the choice. The clearest exception is XPRIZE Rapid Covid Testing, launched in July 2020 in response to the pandemic. Its incidence effect, 3.4 percentage points, is the smallest of the eight cohorts (Appendix Table A15).

With that being said, trends that began earlier would have been visible at the time of design, and XPRIZE could have chased those. Two pieces of evidence suggest otherwise. First, patenting in the prize's technology does not grow faster than in neighbouring technologies before the prize. For this test, we do not restrict ourselves to patents from treatment and control firms. Instead, we use the universe of patents granted by the USPTO from 2015 onwards. These patents provide a clean test for two reasons. They cover the universe of inventions, not a sample of financed firms. Their classification is hierarchical, so technologies can be ordered by distance. The Cooperative Patent Classification (CPC) nests subclasses within classes and classes within sections. For prize p, we define the prize area ($r = 0$) as containing the ten CPC subclasses held by the most participants. Technologies one step removed ($r = 1$) are the other subclasses of the same classes. Technologies two

steps removed ($r = 2$) are the other classes of the same sections. Both outer rings exclude, in every quarter, every subclass that belongs to the area of another XPRIZE competition, to avoid contamination whereby the CPC subclasses for one prize are used as the control subclasses for another prize. Let $n_{pr\tau}$ count the USPTO patents granted in ring r in event quarter τ, and let $y_{pr\tau}$ equal $\ln(1 + n_{pr\tau})$ less its mean over $\tau \in [-12, -1]$.[10] We estimate

$$y_{pr\tau} = \delta_{p\tau} + \beta_0 \mathbf{1}\{r = 0\}\, \tilde{\tau} + \beta_1 \mathbf{1}\{r = 1\}\, \tilde{\tau} + \varepsilon_{pr\tau}, \qquad \tau \in [-12, -1], \tag{4}$$

where $\tilde{\tau} = \tau + 6.5$ centres event time on the pre-launch window. The prize-by-quarter fixed effects $\delta_{p\tau}$ absorb every shock common to the three rings of a prize, such as changes in USPTO examination times. β_0 measures the pre-launch growth of the prize area relative to technologies two steps removed, and $\beta_0 - \beta_1$ its growth relative to technologies one step removed. Multiplying these coefficients by 400 converts log points per quarter into per cent per year. We estimate the same equation over the post-launch quarters $\tau \in [0, 12]$, with $y_{pr\tau}$ and $\tilde{\tau}$ centred on that window, so that pre- and post-launch differences share the same units.

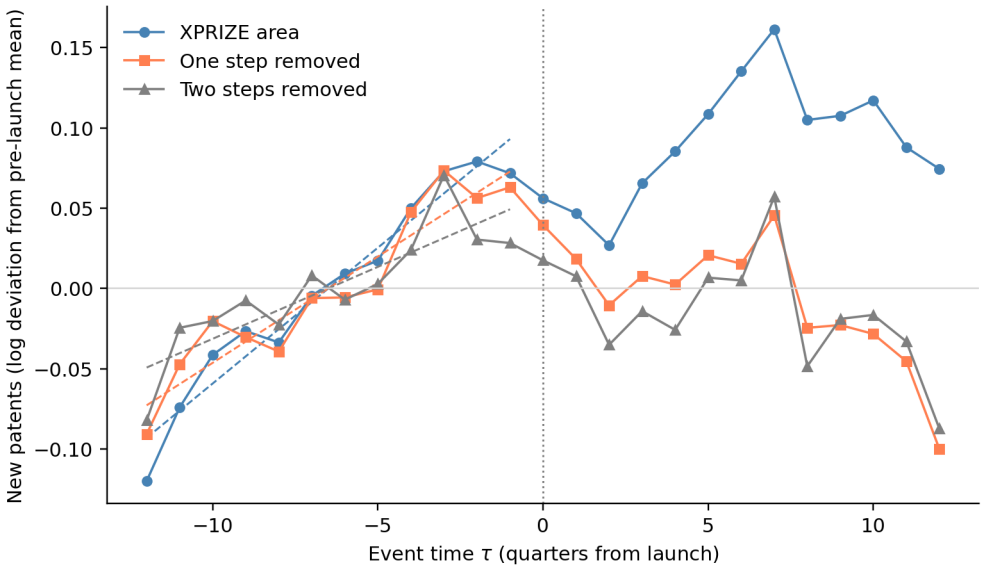
Figure 4 and Table 9 report the results. Each prize receives a weight equal to the pre-launch patents of its smallest ring, so that prizes with thin and noisy series count little.[11] Patenting rises in all three rings before launch. The prize area grows 1.5% per year faster than technologies one step removed, a difference that is not statistically significant. It grows 3.2% per year faster than technologies two steps removed, significant at the 10% level. Weighting prizes by their participant units, in line with the stacked estimator, gives differences of 1.6% and 2.3% per year, with the same significance level. The difference with broader fields comes from XPRIZE Rainforest, whose area includes artificial intelligence and drones, and XPRIZE Rapid Covid Testing. It is close to zero for XPRIZE Carbon Removal and ANA Avatar XPRIZE, which carry the largest weights. After launch, the prize area keeps growing faster than technologies two steps removed, by 3.7% per year (1.1% with participant weights), close to its pre-launch pace. It now also pulls ahead of technologies one step removed, by 4.9% per year (2.9% with participant weights), significant at the 5% level. XPRIZE therefore targets technologies whose inventive activity moves with that of their closest neighbours, and the prize area separates from these neighbours only after

[10] Each patent enters through its primary subclass in the current CPC classification, which assigns every patent under the same scheme whatever its grant date. Patents are dated by their grant quarter. Patents of XPRIZE participants are excluded to isolate the broader technological trends more cleanly. Grant dates are complete from 2015 and the current classification ends in 2024, so the test covers the six prizes launched between 2018 and 2021. These prizes contain 258 of the 383 participant units, including XPRIZE Carbon Removal and XPRIZE Rapid Covid Testing. Appendix A.6 describes the construction.

[11] The noise in $\ln n_{pr\tau}$ falls with $n_{pr\tau}$, and each comparison between two rings is only as precise as the smaller of the two series.

launch. Relative to broader fields, prize areas grow 2 to 3% per year faster throughout.

Figure 4: Patenting in the prize area and in neighbouring technologies



Notes: The figure plots, for each event quarter τ relative to prize launch, the weighted mean of $y_{pr\tau}$ across the six prizes of Appendix Table A4. Each prize is weighted by the mean quarterly pre-launch patents of its smallest ring. $y_{pr\tau}$ is $\ln(1 + n_{pr\tau})$ less its mean over $\tau \in [-12, -1]$, where $n_{pr\tau}$ counts the USPTO patents granted in quarter τ whose primary subclass in the current CPC classification lies in ring r of prize p. The prize area (blue) contains the ten CPC subclasses held by the most participants of the prize. Technologies one step removed (orange) are the other subclasses of the same CPC classes. Technologies two steps removed (grey) are the other classes of the same CPC sections. Both outer rings exclude, in every quarter, the subclasses that form the area of any other XPRIZE competition. Patents of XPRIZE participants are excluded. Dashed lines are least-squares fits over the pre-launch quarters. The vertical line marks the launch quarter $\tau = 0$.

Second, the effect varies little with technological proximity to the prize within the control group. A trend concentrated in the participants' technology would fade with distance from it. Controls closer to the prize would share more of the trend than controls further away, and the effect would be smaller against the former. This test differs from the comparison with distant controls in Section 5.3. That comparison adds a separate group of firms from outside the baseline control group, which share none of the participants' leading keywords, and finds treatment effects slightly above the baseline. Here the participants and the baseline control pool are unchanged, and we compare controls within the pool. Each control receives a proximity score: the share of the participants' keyword tags that fall on keywords the control also carries, so that keywords common among participants weigh more (Appendix C.7). We split the controls of each prize at the median score and estimate the effect against each half. The incidence effect is 7.7 percentage points against controls at or above the median and 7.6 points against those below it (Appendix Table A16).

Table 9: Patenting trends in the prize area and in neighbouring technologies

Contrast	Patent weights	Participant weights
Pre-launch trend difference (% per year)		
Area − one step removed	1.47	1.58
	(1.90)	(1.58)
Area − two steps removed	3.18*	2.26*
	(1.56)	(1.01)
One step − two steps removed	1.71	0.69
	(1.10)	(0.72)
Post-launch trend difference (% per year)		
Area − one step removed	4.93**	2.86**
	(1.80)	(0.82)
Area − two steps removed	3.65	1.09
	(2.08)	(0.93)
One step − two steps removed	-1.27**	-1.77***
	(0.38)	(0.18)
N prizes	6	6
N participant units	258	

Notes: Entries are estimates from Equation (4), with standard errors clustered by prize in parentheses. The outcome $y_{pr\tau}$ follows Figure 4. The upper panel reports differences in pre-launch linear trends, $400(\beta_0 - \beta_1)$, $400\beta_0$, and $400\beta_1$. The lower panel reports the same differences from Equation (4) estimated over $\tau \in [0, 12]$, with $y_{pr\tau}$ and $\tilde{\tau}$ centred on that window. The factor 400 converts coefficients from log points per quarter into per cent per year: 100 converts log points into per cent and 4 converts quarters into years. The first column weights each prize by the mean quarterly pre-launch patents of its smallest ring. The second weights each prize by its participant units in the baseline sample of Table 2. N participant units counts these units across the six prizes. p-values use a t distribution with five degrees of freedom. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The deal-count effect is 0.321 and 0.378 rounds. The count difference of 0.056 rounds is significant at the 5% level. This difference can be attributed to the small spillover effects documented above as well. But even if it reflected a trend entirely, it would amount to one-sixth of the baseline count effect. It is also consistent with the spillovers of Section 5.3, which would raise fundraising most among the controls closest to the prize.

The timing of prize design and the evidence therefore give little support to the view that XPRIZE picks "hot" technologies. The pre-launch rise in fundraising is more consistent with a selection effect of individual firms, which the synthetic control reproduces. Finally, in the context of XPRIZE, self-selection can be part of the signal itself. Entry exposes a firm's technology to independent testing against public criteria, and a firm that doubts its technology has little reason to accept this exposure (Section 2.1).

6 Attracting New Investors

The signalling hypothesis makes a prediction about who responds. A public signal is most valuable to investors who lack private information. A VC that has already financed a firm has studied it closely. A VC that has backed other XPRIZE participants is already familiar with the competition space. A VC that has invested in the prize's technology understands its risks. Investors without such links should respond most to participation. To test this prediction, we shift from counting equity rounds to counting investors. The investor dummy equals one if at least one VC of a given type invests in the firm in quarter τ. The investor count is $\log(1 + x)$ of the cumulative number of distinct VCs of that type. Table 10 draws the line between informed and uninformed investors in three places.

Panel A separates VCs investing in the firm for the first time from VCs that had invested in the same firm. The contrast is stark. Participation raises the quarterly probability that a new VC invests by 3.8 percentage points, but that of a prior VC by only 0.9 points. The number of new VCs rises by 0.128 log points, or 13.7%, whereas the number of prior VCs does not change significantly. Both differences are significant at the 1% level. The additional rounds documented in Section 4 thus bring in investors who had no stake in the firm before, rather than drawing more capital from existing backers.

Panels B and C ask whether these new investors are also new to the prize setting. In Panel B, a prize incumbent had invested in another XPRIZE participant before, and a VC new to XPRIZE had not. Participation raises the probability of investment by VCs new to XPRIZE by 3.0 percentage points, against 1.8 points for prize incumbents. The difference of 1.2 points is significant at the 1% level, although the investor counts of the two groups rise by similar amounts. In Panel C, a VC is new to the technology space if it

Table 10: Equity fundraising from new and incumbent investors

Investor type	Investor dummy	Investor count
Panel A: New to the firm		
New to firm	0.038^{***}	0.128^{***}
	(0.004)	(0.025)
Incumbent investor	0.009^{***}	0.014
	(0.002)	(0.014)
New to firm − Incumbent investor	0.029^{***}	0.114^{***}
	(0.004)	(0.020)
Panel B: New to XPRIZE		
New to Prize	0.030^{***}	0.095^{***}
	(0.004)	(0.022)
Incumbent to prize	0.018^{***}	0.082^{***}
	(0.003)	(0.016)
New to Prize − Incumbent to prize	0.012^{***}	0.013
	(0.004)	(0.019)
Panel C: New to the technology space		
New to space	0.027^{***}	0.104^{***}
	(0.003)	(0.021)
Incumbent to space	0.021^{***}	0.065^{***}
	(0.003)	(0.016)
New to space − Incumbent to space	0.007^{*}	0.039^{*}
	(0.004)	(0.020)
N treated firms	383	383
N treated equity deals	823	823

Notes: Entries are stacked SDID estimates of the treatment effect, with standard errors from 50 unit-bootstrap replications in parentheses. Outcomes and support rows follow Table 2. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The investor dummy equals one if at least one VC investor of the stated type invests in the firm in the quarter. The investor count is $\log(1 + x)$ of the cumulative number of distinct VC investors of that type. Panel A: "New to firm" denotes a VC's first investment in the firm; "Incumbent investor" denotes a VC with an earlier investment in the same firm. Panel B splits VCs new to the firm: "New to Prize" denotes no earlier investment in another XPRIZE participant before the focal investment; "Incumbent to prize" denotes at least one. Panel C splits VCs new to the firm by prior technology exposure: "New to space" denotes no earlier investment in a firm tagged with any of the prize's ten leading participant keywords; "Incumbent to space" denotes at least one. Difference rows use paired-bootstrap standard errors.

has never invested in a firm tagged with any of the ten leading keywords of the prize's participants. These entrants again respond more than incumbents, by 0.7 percentage points in incidence and 0.039 log points in count. Both differences are significant at the 10% level. With a broader definition of the prize technology space, using the top twenty keywords, the entrant–incumbent differences retain their positive sign but attenuate and are no longer statistically significant (Appendix Table A20). As we expand the delimitation of the technology spaces with keywords less central to the prize, the treatment effects weaken, suggesting that the prize signal is more credible for its core technology areas. Furthermore, we test if the xprize helps firms outside the U.S raise money from U.S based investors and *vice versa*. We find no such effect. The results shown in appendix E.2 suggest that, if anything, Xprize reinforces the probability that US investors invest in US firms and non-US investors in non-US firms. This result suggest that the xprize alleviates technological uncertainty for investors, more so than search frictions that may prevent US investors to invest in non-US firms for example.

Read together, the three panels suggest that participation draws in VCs with less prior exposure to the firm, to XPRIZE, and to the prize's technology. The entrant advantage is clearest under the narrower ten-keyword definition and attenuates under the broader definition. This pattern is consistent with the prize signal attracting investors towards less familiar technology areas.

7 Grant Funding

So far we have studied private investors. Equity is not the only source of finance for early-stage deep-tech firms. Grants are a major alternative. Before launch, 20.9% of participants had received a grant, more than the 19.0% that had raised a VC round (Table 1). Across the PitchBook deals of participant firms, grants number 531 against 651 VC rounds, and they account for 531 of the 603 non-dilutive deals (Appendix Figure A1). Most come from public funders of science and technology, led by the National Science Foundation, the US Department of Energy, NASA, and the US Department of Defense, with Innovate UK the leading non-US provider (Appendix Table A7). These grants matter beyond their amount: early-stage awards raise the probability that a firm later obtains venture capital (Howell, 2017; Lerner, 2000).

How public funders respond to a prize is not obvious, because two views of their role give opposite predictions. The first casts public agencies as lead investors that absorb technological uncertainty before private capital enters (Azoulay et al., 2019a; Mazzucato, 2018; Mazzucato and Semieniuk, 2017). Agencies that resolve technological uncertainty

through their own technical review have no use for an external signal, so participation should leave grant funding unchanged. It could even reduce it, if agencies view participants as already supported by the competition or stretched by its demands. The second view holds that public funders, despite their mission of funding breakthrough science, can be conservative. Public funders sometimes penalise novel and uncertain proposals (Azoulay, Graff Zivin and Manso, 2011; Azoulay and Greenblatt, 2025; Boudreau et al., 2016; Franzoni, Stephan and Veugelers, 2022). Public R&D competitions that specify the desired product also attract fewer start-ups than open ones (Howell et al., 2023). Agencies that avoid unproven technologies should value an external validation, and participation should raise grant funding as it raises equity. Table 11 tests these predictions with the same two outcomes as for equity: the grant dummy and the cumulative grant count.

Public funders do respond. Participation raises the quarterly probability of a grant by 1.2 percentage points and the cumulative grant count by 0.117, both significant at the 1% level. The response is, however, much weaker than for equity. The second panel of Table 11 shows that the equity effect exceeds the grant effect by 6.5 percentage points in incidence and by 0.221 in count, both significant at the 1% level.

Table 11: Grant funding

Estimate	Grant dummy	Grant count
Grant effect		
ATT	0.012^{***}	0.117^{***}
	(0.004)	(0.023)
N treated firms	383	383
N treated deals	286	286
Equity effect − grant effect		
ATT	0.065^{***}	0.221^{***}
	(0.006)	(0.054)

Notes: Entries are stacked SDID estimates of the treatment effect, with standard errors from 50 unit-bootstrap replications in parentheses. Outcomes and support rows follow Table 2. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The grant dummy equals one if the firm receives at least one PitchBook grant in the quarter. The grant count is the cumulative number of grants since $\tau = -12$, in levels. Grants for which XPRIZE is the provider are excluded. N treated deals are participant grants within the event window. The second panel reports the equity effect (Table 2) less the grant effect for each displayed outcome, with standard errors from paired bootstrap draws on the same resampled units.

The prize thus matters for both sources of early-stage finance, but far more for private equity. The evidence lies between the two views in the literature. Public funders respond to participation, in contrast to what their mission of independently resolving technological uncertainty would suggest. Their response is, however, one-sixth of the equity response in

incidence and one-third in count. This gap is consistent with the view that public agencies are more willing to invest in uncertain, unproven technologies and therefore less reliant on an external signal than private investors are.

8 Conclusion

Innovation prizes are usually studied as incentives: a reward that induces innovative effort towards a defined goal. This paper shows that prizes serve a second function. They signal quality to investors and thereby shape the financing of competing firms. Matching XPRIZE competition data to PitchBook records and using a synthetic difference-in-differences design, we find that prize participation raises the quarterly probability of raising equity by 7.7 percentage points, a 164% increase on the pre-prize level. The effect persists for at least three years and, averaged over that period, adds one-third of an equity round per participant. Participation also raises equity deal sizes and valuations, although data on these two outcomes are more limited.

The pattern of the effect points to information as the channel. Participation signals the quality and potential of a firm's technology to investors. Consistent with this interpretation, the effect is strongest among firms that lack other quality signals: it is twice as large for firms without a patent as for firms with one, and four times as large for firms with neither a patent nor prior equity as for firms with at least one of the two. The effect also grows with the size of the competition and the monetary value of its prize, which suggests that higher-profile competitions send stronger signals. Finally, participation mainly attracts investors who are new to the firm, new to XPRIZE and new to the technology space, again suggesting information provision as the primary mechanism: The investors who respond the most or the new entrants with the least prior exposure to the technology space and therefore the greatest need for outside information. The signal accrues mostly to participants and extends little to non-participants operating in a similar technology space. By demonstrating the promise of a technology space, XPRIZE competitions do raise financing for neighbouring firms. But this spillover is only about one-tenth of the direct effect on participants for the probability of raising equity and one-fifth for the number of equity rounds.

The signal also reaches public funders. Many of these funders, such as the US Department of Energy or the Department of Defense, focus on funding nascent technologies that carry considerable technological uncertainty. That they respond to the prize at all is perhaps surprising. Their response is nonetheless far weaker than that of private investors: their treatment effect is roughly a quarter to a sixth of the effect on private investors.

These findings point to a further role for innovation prizes. Beyond rewarding a winner and inducing innovative effort, a prize produces public information on the viability of the technologies it targets and on the potential of the firms that develop them. This information can reduce the technological uncertainty that has historically hampered early-stage firms in deep-tech sectors such as energy or space.

Future work could identify which features of innovation prizes generate the most information. Is it the independently defined metrics that assess a firm's technology, both in absolute terms and relative to other participants? Or is it simply the media visibility that a prize confers on participants? Answering these questions would help design future prizes that crowd in more private investment.

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A Data and Sample Construction

A.1 Registered and qualified teams

Table A1 reports, for each of the 25 competitions, the teams that completed registration and the teams that qualified after the XPRIZE review. Of 5,309 registered teams, 1,172 qualified (22.1%). The median competition qualified 30.3% of its registered teams. The pooled share is lower because the competitions with the most registrations are selective: each of the six competitions with more than 300 registered teams qualified at most 30.3% of them. XPRIZE Carbon Removal qualified 336 of 1,334 teams (25.2%), XPRIZE Feed the Next Billion 32 of 372 (8.6%), and the Next-Gen Mask Challenge 28 of 991 (2.8%).

Table A1: Registered and qualified teams by competition

		Teams		
Prize	Period	Registered	Qualified	Qualified (%)
Suborbital Flight	1996–2004	26	5	19.2
Lunar Lander	2006–2009	10	2	20.0
Lunar	2007–2018	31	20	64.5
Automotive	2009–2010	111	13	11.7
Oil Cleanup	2010–2011	27	10	37.0
Tricorder	2012–2017	11	11	100.0
Sensing	2012–2014	67	23	34.3
Ocean Health	2013–2015	23	15	65.2
Global Learning	2014–2019	8	8	100.0
Adult Literacy	2015–2019	24	17	70.8
Carbon	2015–2021	26	25	96.2
Ocean Discovery	2015–2019	19	19	100.0
AI	2016–2021	458	68	14.8
Women's Safety	2016–2018	85	21	24.7
Water Abundance	2016–2018	98	6	6.1
Avatar	2018–2022	111	84	75.7
Rainforest	2019–2024	118	34	28.8
Rapid Reskilling	2020–2023	118	12	10.2
Next-Gen Mask	2020	991	28	2.8
Rapid COVID Testing	2020–2021	722	219	30.3
Pandemic Response	2020–2021	368	104	28.3
Feed the Next Billion	2020–2024	372	32	8.6
Digital Learning	2021–2023	33	10	30.3
Carbon Removal	2021–2025	1,334	336	25.2
Wildfire	2023–2026	118	50	42.4
Total		5,309	1,172	22.1

Notes: XPRIZE Foundation records, extracted on 28 September 2026. Competitions are ordered by launch date. Prize labels omit sponsor names and the XPRIZE/XCHALLENGE designation. Period reports the launch and end years of the competition. Registered counts teams that completed XPRIZE registration. Qualified counts registered teams that passed the XPRIZE review of their organisation and technical approach and were admitted as official competitors. Qualified (%) divides qualified by registered teams. The total row pools teams across competitions. The table omits Genomics XPRIZE, which has no registration records, and the Adult Literacy XPRIZE Communities Competition.

A.2 Panel construction

The participant sample starts from the XPRIZE team roster. XPRIZE records a PitchBook company identifier for each team that could be linked to a PitchBook firm. Teams without a PitchBook identifier are not part of the estimation sample. We exclude firms founded before 1990. Firms with a missing founding year remain eligible.

The unit before stacking is the firm–quarter. A deal is dated by its PitchBook deal date, or by its announcement date when the deal date is missing. Several deals can occur for one

firm in one quarter. We take the maximum of binary deal indicators, count the underlying deals, and sum deal amounts. The firm panel starts in the earlier of the firm's first deal quarter and its prize launch quarter. It ends at the first recorded exit quarter or in 2025Q2. Missing quarters between these endpoints are inserted with zero deal indicators.

The event-time panel keeps, for each firm, the 25 quarters $\tau = t - L_p \in [-12, 12]$, where L_p is the launch quarter of prize p. Units without the complete window are excluded. Quarters before a firm's first recorded deal are set to zero for participants and controls alike. The two amount measures of Appendix D rest on an audit of every in-window equity deal. Among the 894 in-window equity deals of all matched participants, 402 (45.0%) report no amount. The total-recorded measure sets these amounts to zero. The first-recorded measure retains only units whose first equity round with a reported amount falls within the window.

Table A2 compares finalist and non-finalist equity fundraising in the five largest participant cohorts with at least one finalist in the balanced estimation sample. These are descriptive differences: finalist status is determined after launch.

Table A2: Equity fundraising in the five largest prizes with finalists in the sample

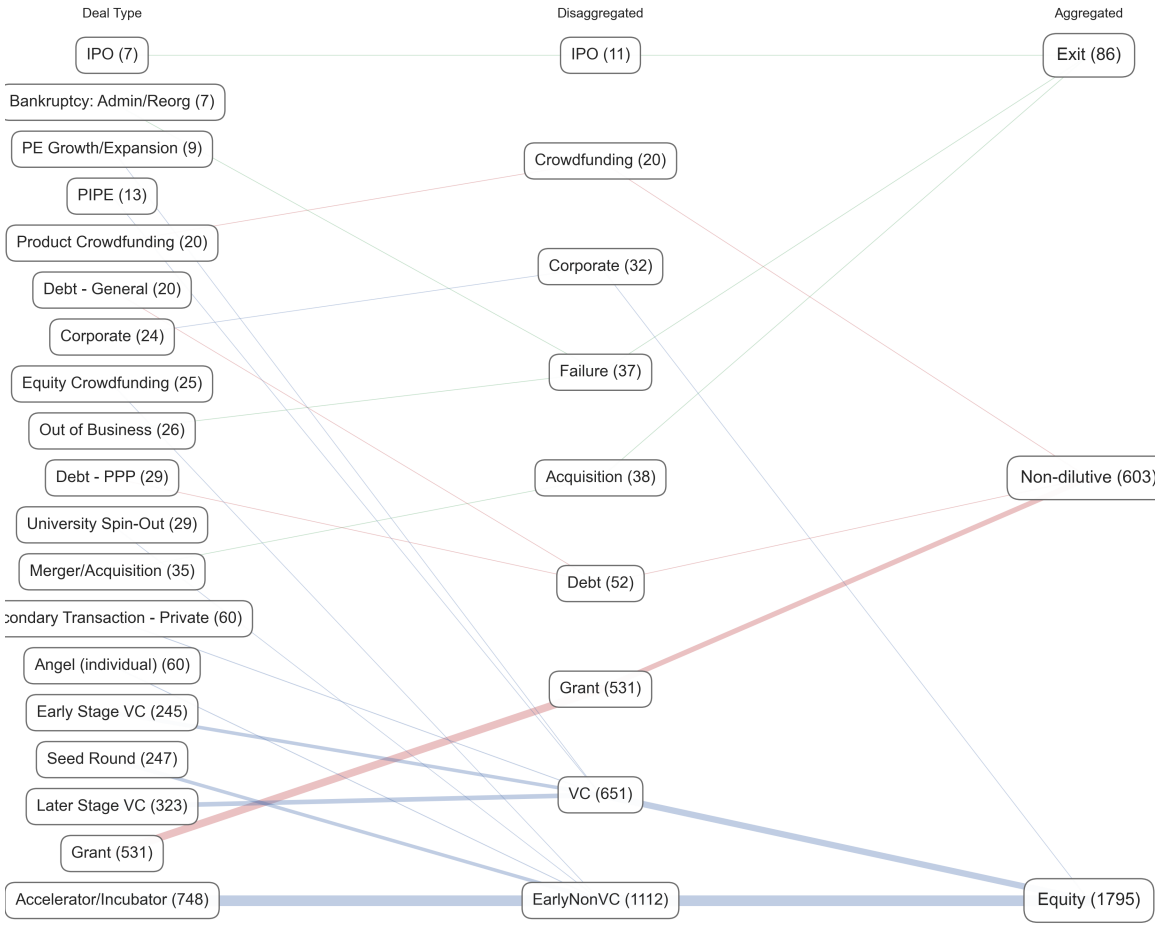
	Finalists				Non-finalists			
	Equity dummy (%)		Equity deal count		Equity dummy (%)		Equity deal count	
Prize	Pre / post	% change	Pre / post	% change	Pre / post	% change	Pre / post	% change
Rapid Covid Testing	3.2 / 10.1	+213.8	1.269 / 2.130	+67.8	6.3 / 7.3	+16.3	1.327 / 2.324	+75.1
N firms	*13*		*13*		*66*		*66*	
IBM Watson AI	12.5 / 15.4	+23.1	0.292 / 2.115	+625.3	4.3 / 10.9	+153.8	0.272 / 1.536	+465.7
N firms	*2*		*2*		*31*		*31*	
ANA Avatar	0.0 / 9.6	–	0.000 / 0.288	–	4.4 / 11.3	+158.5	0.268 / 1.543	+476.5
N firms	*4*		*4*		*19*		*19*	
NRG Cosia Carbon	5.3 / 11.2	+111.0	0.515 / 1.706	+231.2	3.7 / 9.4	+153.8	0.185 / 1.179	+536.9
N firms	*11*		*11*		*9*		*9*	
Nokia Sensing	1.8 / 4.4	+146.2	0.339 / 0.731	+115.4	5.6 / 7.7	+38.5	0.917 / 2.026	+121.0
N firms	*14*		*14*		*3*		*3*	

Notes: Prizes are ordered by participant units in the balanced sample of Figure 3. Finalists include winners. The italicised second row labels the firm counts; counts beneath each pre/post mean give the pre- and post-period firms, respectively. For each group and outcome, the first row reports the pre-launch mean ($\tau \in [-12, -1]$) / post-launch mean ($\tau \in [0, 12]$), followed by the relative percentage change, calculated as $100(\bar{Y}_{post}/\bar{Y}_{pre} - 1)$. The equity dummy is the quarterly probability of raising an equity round, in per cent; the deal count is cumulative from $\tau = -12$. Percentage changes are undefined when the pre-launch mean is zero and are shown as "–". Carbon Removal and Feed the Next Billion are not shown because neither has a finalist in the balanced estimation sample, although each raw XPRIZE roster flags one. The Carbon Removal finalist, ACIDD PROJECT, has no PitchBook match (its PB ID is "Not Found"). The Feed the Next Billion finalist, Kernel Mycofoods, has a valid PB ID, but PitchBook records its acquisition in 2021Q4; its panel therefore lacks the complete 25-quarter event window.

A.3 Deal classification

Figure A1 records the mapping from PitchBook deal types to the categorisation used in the analysis. The raw deal types map to nine groups: IPO (11), crowdfunding (20), corporate (32), failure (37), acquisition (38), debt (52), grant (531), VC (651), and early non-VC (1,112). These groups then map to three mutually exclusive classes: exit (86), non-dilutive (603), and equity (1,795). Equity combines VC, early non-VC, and corporate rounds. The grant outcomes of Section 7 retain only deals labelled as grants and exclude grants provided by XPRIZE.

Figure A1: Classification of PitchBook deal types



Notes: The left column reports raw PitchBook deal types and their frequencies. The middle column reports the disaggregated groups. The right column reports the three classes used in the analysis. Line width is proportional to the number of deals mapped between categories. Counts refer to deals or exit events, not to firms.

A.4 Control pools

For prize p, let $\mathcal{T}_p$ denote its participants, K_i the PitchBook keyword set of firm i, and $n_p(w)$ the number of participants tagged with keyword w. A non-participant is a candidate control for prize p if K_i contains at least one of the twenty keywords with the largest $n_p(w)$ and the firm records a deal in the five years before launch (any deal not necesserily equity). Its technological proximity to the prize is

$$\pi_{ip} = \frac{\sum_{w \in K_i} n_p(w)}{\sum_{j \in \mathcal{T}_p} |K_j|}. \tag{5}$$

A keyword shared by many participants therefore contributes more to proximity. A control eligible for several prizes is assigned only to the prize with the highest π_{ip}, so that no control enters two pools. Prizes with fewer than ten matched participants form the "Other" cohort. Google Lunar XPRIZE also joins this cohort because its distant control ring is too small for prize-level estimation. Within "Other", every firm keeps its own prize identity and launch date.

The distant ring of Section 5.3 uses the twenty keywords most frequent among a prize's adjacent controls, after removing the participants' keywords. A distant firm shares none of the participants' twenty leading keywords, shares at least one of these adjacent keywords, and records a deal in the five years before launch. The proximity split of Appendix C.7 divides adjacent controls at the within-prize median of π_{ip}.

A.5 Keyword Controls

Table A3 lists the twenty leading participant keywords for the four largest participant cohorts. Figure A2 compares representative adjacent controls from the upper and lower quartiles of within-prize keyword proximity. The lower-quartile firms are not members of the distant ring defined above.

Table A3: Leading participant keywords in four XPRIZE competitions

XPRIZE Carbon Removal
carbon technology; carbon tech; directing air capture; carbon removal; carbon capture; biological carbon removal; clean fuel; climate change; agriculture technology; carbon dioxide removal; renewable energy; green construction; carbon credit; built environment; climate technology; environmental services; carbon removal technology; ag biotech; plant biotechnology; carbon sequestration
XPRIZE Rapid Covid Testing
rapid testing; point-of-care testing; medical diagnostics; medical device; diagnostic technology; rapid test; health monitoring; molecular diagnostics; molecular testing; medical technology; ag biotech; agriculture technology; diagnostic platform; covid-19 test; diagnostic device; food safety; molecular biology; biotechnology research; diagnostic tools; diagnostic test
IBM Watson AI XPRIZE
artificial intelligence; enterprise resource planning; animal agriculture; vertical application; natural language technology; machine learning; medical equipment; learning platform; educational platform; engine reconstruction technology; artificial intelligence app; distracted driving; driving performance; fleet connectivity; fleet management; fleet management software; freight technology; mapping technology; road safety; surveillance technology
XPRIZE Feed the Next Billion
alt-proteins; cultivated protein; cellular agriculture; plant-based protein; fermented protein; clean meat; cultivated meat; plant-based meat; alternative protein; sustainable food; meat alternative; food technology; plant protein; seafood product; cultured meat; plant meat; cell-based meat; sustainable seafood; meat substitute; nutritional food

Notes: PitchBook keywords assigned to participant firms, ordered by the number of participants tagged with each keyword. These twenty keywords define the adjacent control pool for each prize; keyword counts are omitted for space.

Figure A2: Firm descriptions at high and low keyword proximity

XPRIZE Carbon Removal

Top 25%
π=0.023; $\bar{k}$=2.1; n=454

Developer of synergistic technologies designed to convert wet organic and plastic wastes into green hydrogen. The company's technology converts organic and plastic waste into green hydrogen, reducing the environmental burden of waste disposal and providing a sustainable energy source...

Bottom 25%
π=0.008; $\bar{k}$=1.0; n=454

Provider of novel bio-herbicides intended to exploit sterility against resistant weeds. The company's herbicides are made from weed pollen, which is applied during the weed flowering period via artificial pollination, which prevents the generation of viable resistant weed seeds, enabling farmers and...

XPRIZE Rapid Covid Testing

Top 25%
π=0.010; $\bar{k}$=1.4; n=184

Developer of an antibiotic susceptibility testing system designed to provide live imaging of cell behaviour in response to defined concentration gradients. The company's system combines microfluidics with live imaging to analyze bacterial responses on a linear antibiotic concentration...

Bottom 25%
π=0.003; $\bar{k}$=1.0; n=184

Provider of biotechnology research and development services based in Belfast, Northern Ireland. The company specializes in Research and experimental development in biotechnology and offers Information Technology Services, biotechnology research and development, and Human Health Services.

IBM Watson AI XPRIZE

Top 25%
π=0.018; $\bar{k}$=1.3; n=89

Veritone Inc is a cloud-based cognitive software company. Its proprietary artificial intelligence platform unlocks the power of cognitive computing to transform unstructured audio and video data and analyze it in conjunction with structured data in a seamless, automated manner to...

Bottom 25%
π=0.004; $\bar{k}$=1.0; n=89

Developer of a communications intelligence platform designed to unlock the power of conversational data for customers and enterprises. The company's platform utilizes artificial intelligence and machine learning to provide real-time, structured annotations of the meaning and emotion contained in...

XPRIZE Feed the Next Billion

Top 25%
π=0.079; $\bar{k}$=2.4; n=60

Manufacturer of dairy alternative food products based in Burnaby, Canada. The company's line of products includes shreds, slices, vegetable crust and gluten-free pizza, flatbreads, mac and cheese, cheesecake and dairy-free dressings, thereby enabling its clients by offering custom food...

Bottom 25%
π=0.014; $\bar{k}$=1.0; n=60

Producer of food products intended to upcycle crop leftovers into healthy and sustainable ingredients. The company uses proprietary technology that can integrate seamlessly with the traditional agricultural system to convert agricultural residues into quality products, enabling farmers to...

Notes: Each panel shows a PitchBook description from the top and bottom quartiles of eligible adjacent controls, selecting the described firm closest to each quartile's median proximity. π is the selected firm's score (Equation (5)); $\bar{k}$ is the mean number of leading participant keywords shared by controls in that quartile, and n is the number of controls in the quartile. Distant controls share none of the prize's twenty leading participant keywords.

A.6 Patent markets

The patent test of Section 5.4 draws on two PatentsView files. The current CPC file classifies every patent granted by the USPTO under the latest CPC scheme. Revisions of the scheme regularly carve new subclasses out of existing ones. The current file reassigns earlier patents to the new subclasses, so that a technology keeps one code whatever the grant date of its patents. The at-issue CPC file records the grant date of every patent granted from mid-2013, and its coverage is complete from 2015. The current file ends with patents granted in 2024Q4. Each patent enters once, through its first-listed subclass in the current classification, and is dated by its grant quarter. We remove the 829 patents matched to XPRIZE participants through the PitchBook–USPTO assignee match provided by PitchBook.

The prize area comes from the CPC field of the prize's participants in PitchBook. We match each entry of this field to the CPC subclass whose official title begins with the same clause. The prize area contains the ten subclasses held by the most participants. Ties are broken by the number of participant patents in the subclass and then by CPC order. XPRIZE Rapid Reskilling has only three participant subclasses. The CPC hierarchy then defines the two outer rings: the other subclasses of the area's classes, and the other classes of the area's sections. The same procedure yields an area for 19 of the 25 XPRIZE competitions, and these areas cover 80 subclasses in total. The outer rings of a prize exclude every subclass in the area of another competition. The exclusion is fixed: these subclasses leave the outer rings in every quarter, before and after the launch of any prize. A neighbouring technology can therefore never be the target of another prize, and the composition of the rings does not change over the event window. A prize enters the test if it has at least one area subclass and its window $\tau \in [-12, 12]$ lies between 2015Q1 and 2024Q4. Earlier prizes lack complete pre-launch coverage. Next-Gen Mask Challenge, Pandemic Response Challenge, and Digital Learning Challenge have no participant with a CPC title. XPRIZE Wildfire's window extends beyond 2024Q4. Six prizes remain.

Table A4 lists their area subclasses, the number of subclasses removed from their outer rings, and the mean number of patents granted per quarter in each ring before launch. All rings but one receive more than 1,000 patents per quarter, so that their quarterly counts are smooth. The exception is the ring one step removed of XPRIZE Rapid Reskilling, with 286 patents per quarter: most other subclasses of its class G06 belong to the areas of other prizes. Section 5.4 weights each prize by the pre-launch patents of its smallest ring, which gives this prize a weight of 1.2%. Table A4 reports the patent and participant weights of every prize.

Table A4: Patent markets of the six prizes

Prize	Launch	Area subclasses	Excluded subclasses	Pre-launch patents per quarter			Weight (%)	
				Area	One step removed	Two steps removed	Patents	Participants
Avatar	2018Q1	A41D; A61B; A61N; A63F; B06B; B25J; G01L; G06F; H02K; H04R	48	11,427	10,234	21,698	43.5	8.9
Rainforest	2019Q4	B01D; B64C; B64U; C12N; G01N; G05D; G06N; G06Q; G06T; G06V	45	5,766	3,084	13,082	13.1	3.1
Rapid Reskilling	2020Q2	G06F; G06Q; G06T	14	10,752	286	7,277	1.2	1.6
Rapid COVID Testing	2020Q3	A61B; A61K; B01L; C07K; C12M; C12N; C12Q; G01N; G06F; G16H	55	15,909	6,002	16,206	25.5	30.6
Feed the Next Billion	2020Q4	A01G; A21D; A23J; A23L; A23P; B01J; B01L; B05B; C12M; C12N	38	1,407	1,386	13,335	5.9	10.5
Carbon Removal	2021Q2	B01D; B01J; B09B; C01B; C01F; C02F; C04B; C05F; C10B; G01N	45	2,651	2,541	14,228	10.8	45.3

Notes: Launch is the launch quarter of the prize. Prize labels omit sponsor names and the XPRIZE designation. Area subclasses are the ten CPC subclasses held by the most participants in PitchBook. The "Excluded subclasses" column counts the area subclasses of other XPRIZE competitions that lie in the CPC sections of the prize; these subclasses are removed from both outer rings. The last three columns report the mean number of USPTO patents granted per quarter over $\tau \in [-12, -1]$ whose primary subclass in the current CPC classification lies in the prize area, in the other subclasses of the same CPC classes (one step removed), and in the other classes of the same CPC sections (two steps removed), after the exclusion. Patents of XPRIZE participants are excluded. The patent weight of a prize is proportional to the pre-launch patents of its smallest ring; the participant weight is proportional to its participant units in the baseline sample of Table 2.

A.7 Prize purse measures

The expected purse per participating team of prize p is P_p/R_p, where P_p is the headline purse in real 2017 USD and R_p is the number of unique, non-excluded teams in the XPRIZE team roster. The denominator therefore counts all participating teams, including those without a PitchBook match. The roster is also the source of the participant sample (Appendix A.2). It is distinct from the registration records of Table A1, so R_p can differ from the qualified count reported there. Tables A5 and A6 list every prize on each side of the USD 0.200 million expected purse threshold. The lower group contains 629 participating teams across eleven prizes and the upper group 651 across fourteen prizes. XPRIZE Carbon Removal has the largest purse, USD 90.79 million, but lies close to the threshold at an expected purse of USD 0.258 million per team because it has 352 participating teams.

Figure A3 shows why the two purse measures differ. Across the 25 prizes, the number of participating teams rises with the headline purse, with a slope of 3.074 teams per USD 1 million (standard error 0.589) and $R^2 = 0.542$. Large prizes thus attract more competitors, and the expected prize per team rises less than proportionally with the purse.

Table A5: Prizes at or below the expected purse threshold

Prize	Participating teams	Purse (2017 USD m)	Expected purse per team (2017 USD m)
Below expected purse threshold (at most 0.200 2017 USD m per team)			
Pandemic Response	106	0.47	0.004
Rapid COVID Testing	211	5.70	0.027
Next-Gen Mask	29	0.95	0.033
Women's Safety	21	1.02	0.048
AI	73	5.09	0.070
Digital Learning	11	0.91	0.083
Sensing	23	2.41	0.105
Avatar	90	9.78	0.109
Ocean Health	19	2.11	0.111
Oil Cleanup	10	1.56	0.156
Adult Literacy	36	7.19	0.200
Below-boundary participating teams	629		

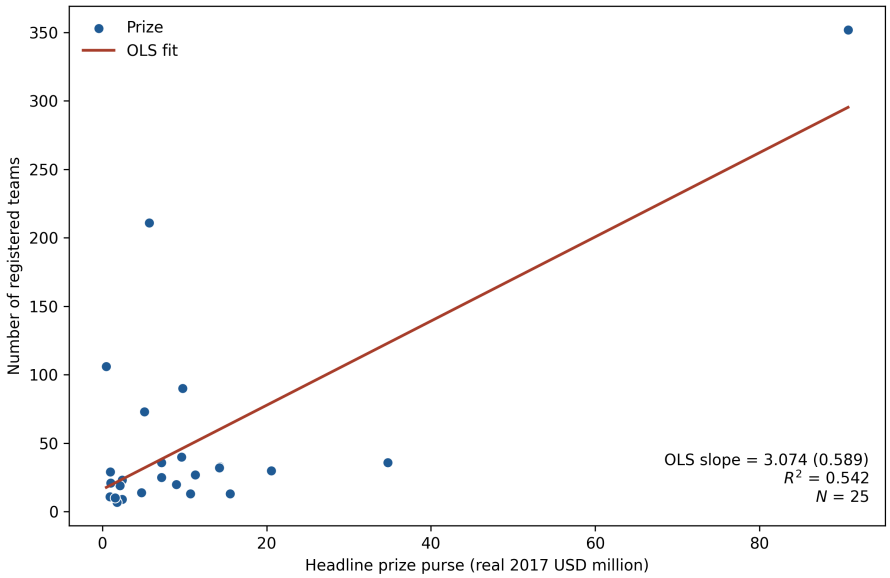
Notes: Participating teams are the unique, non-excluded teams in the XPRIZE team roster for each prize. Purse is the headline prize purse in real 2017 USD millions. Expected purse per team divides the purse by participating teams. The group contains prizes with an expected purse of at most USD 0.200 million per team.

Table A6: Prizes above the expected purse threshold

Prize	Participating teams	Purse (2017 USD m)	Expected purse per team (2017 USD m)
Above expected purse threshold (more than 0.200 2017 USD m per team)			
Rainforest	40	9.62	0.240
Water Abundance	7	1.78	0.254
Carbon Removal	352	90.79	0.258
Lunar Lander	9	2.38	0.264
Ocean Discovery	25	7.19	0.288
Rapid Reskilling	14	4.75	0.339
Automotive	27	11.29	0.418
Suborbital Flight	33	14.26	0.432
Feed the Next Billion	32	14.24	0.445
Wildfire	20	9.00	0.450
Carbon	30	20.55	0.685
Tricorder	13	10.73	0.825
Lunar	36	34.74	0.965
Global Learning	13	15.56	1.197
Above-boundary participating teams	651		

Notes: Definitions follow Table A5. The group contains prizes with an expected purse of more than USD 0.200 million per team.

Figure A3: Headline prize purse and participating teams



Notes: Each point is one of the 25 prizes used for the purse-per-participating-team split. The horizontal axis reports the headline purse in real 2017 USD millions. The vertical axis reports unique, non-excluded participating teams. The line is an ordinary least-squares fit; the slope standard error is in parentheses.

A.8 Public funders

Table A7 ranks the ten leading public grant providers by the number of distinct grants to participant firms. The providers are predominantly public science and technology funders. The National Science Foundation funded the most participant firms (34), while the Department of Energy supplied the most grant capital (USD 186.41 million).

Table A7: Leading public grant providers to participant firms

Rank	Grant provider	Participant firms	Distinct grants	Total grant capital ($m)	Median grant ($m)
Public grant providers					
1	National Science Foundation	34	63	61.21	0.24
2	U.S. Department of Energy	26	47	186.41	1.51
3	National Aeronautics and Space Administration	9	30	79.19	0.80
4	United States Department of Defense	18	26	62.99	0.92
5	U.S. Department of Health and Human Services	17	26	22.91	0.41
6	Innovate UK	11	20	14.03	0.33
7	National Institutes of Health	9	17	25.12	0.60
8	National Institute for Allergy and Infectious Disease	5	11	18.67	1.92
9	United States Department of Agriculture	7	10	15.06	0.92
10	Eunice Kennedy Shriver National Institute of Child Health and Human Development	2	8	19.66	2.12

Notes: Records cover participant firms through 2025Q2. Grant providers are ranked by the number of distinct grants to participant firms. Participant firms counts distinct firms receiving a grant from the provider. Distinct grants counts provider–grant relationships. Total grant capital is in real 2017 USD millions; the full reported amount is attributed to each provider, so provider totals are not additive. Median grant is the median reported grant amount among grants involving the provider. Medians exclude grants without a reported amount.

B Synthetic Difference-in-Differences Details and Diagnostics

B.1 Stacking, cohort weighting, and inference

The stacked estimator of Section 3.1 applies Equation (3) once to the panel of all aligned cohorts. It pools common event quarters, while each unit keeps its own prize assignment. An alternative estimates $\widehat{\theta}_p$ separately for each raw prize and aggregates,

$$\widehat{\theta}_w = \sum_p \frac{w_p}{\sum_q w_q} \widehat{\theta}_p. \tag{6}$$

Standard cohort weighting sets $w_p = N_{1p}T_{1p}$, the number of participant unit–quarters after launch. Inverse-purse weighting sets $w_p = N_{1p}T_{1p}/P_p$, which divides each cohort's weight by its headline purse P_p. Inverse-expected-prize weighting sets $w_p = N_{1p}T_{1p}/(P_p/R_p)$, where R_p is the number of participating teams. The two alternatives shift weight towards prizes with small purses and small expected prizes. Appendix Table A12 reports the three weights for each prize. The standard cohort-weighted estimate need not equal the stacked estimate. It fits separate weights within each prize, and it retains only the 18 raw prizes whose cohorts can be estimated separately.

Let $s_p = N_{1p}T_{1p}/\sum_q N_{1q}T_{1q}$ denote the standard weight share of prize p, and let $\theta_p = \alpha + \beta P_p$. Standard weighting gives

$$\theta_w = \sum_p s_p \left(\alpha + \beta P_p\right) = \alpha + \beta \bar{P}, \qquad \bar{P} = \sum_p s_p P_p. \tag{7}$$

Inverse-purse weighting sets $w_p \propto s_p/P_p$ and gives

$$\theta_w = \frac{\sum_p \left(s_p/P_p\right)\left(\alpha + \beta P_p\right)}{\sum_q s_q/P_q} = \alpha + \beta\Big(\sum_p \frac{s_p}{P_p}\Big)^{-1} = \alpha + \beta \tilde{P}^H. \tag{8}$$

$\bar{P}$ is the arithmetic mean purse across participant unit–quarters and $\tilde{P}^H$ its harmonic mean. The shares of Appendix Table A12 give $\bar{P} = 34.7$ and $\tilde{P}^H = 6.5$ USD million. The harmonic mean is dominated by the smallest purses, so inverse-purse weighting evaluates the purse gradient near a small prize. Inverse-expected-prize weighting applies the same argument to P_p/R_p. If θ_p is linear in the expected prize, it evaluates the gradient at a harmonic mean of USD 0.063 million, against an arithmetic mean of USD 0.220 million.

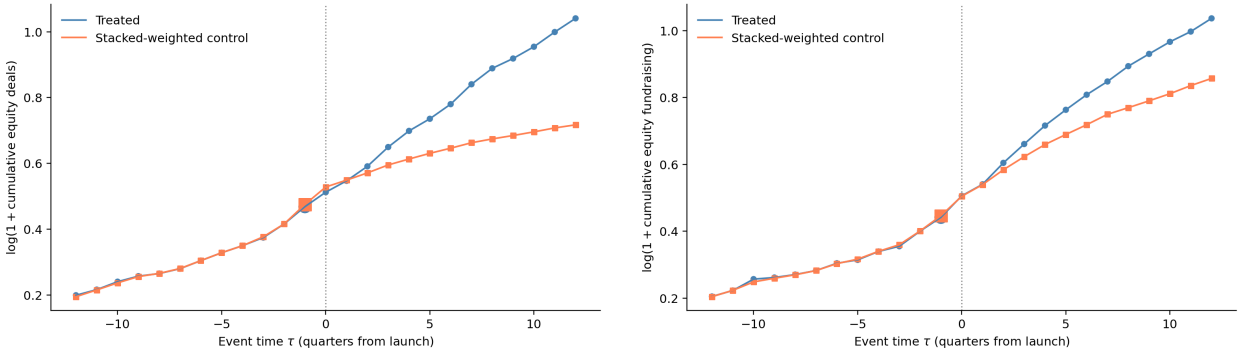
Inference resamples complete firm–prize histories with replacement. Each draw relabels duplicated units, re-estimates both sets of weights, and recomputes the estimate. Draws containing only one treatment status are replaced. For a contrast between two estimates,

both are recomputed within the same draw, and the standard error is the standard deviation of the within-draw differences. No outcome reuses the weights estimated for another outcome.

B.2 Pre-launch fit

Figures A4(A) and A4(B) show the pre-launch fit of the equity deal count and total recorded equity amount side by side. Figure A5 reports the alternative amount samples. Figure A6 reports the fit of the investor outcomes in Panels A–C of Table 10. In each case, the participant and synthetic-control paths coincide before launch.

Figure A4: Pre-launch fit: equity deal count and amount

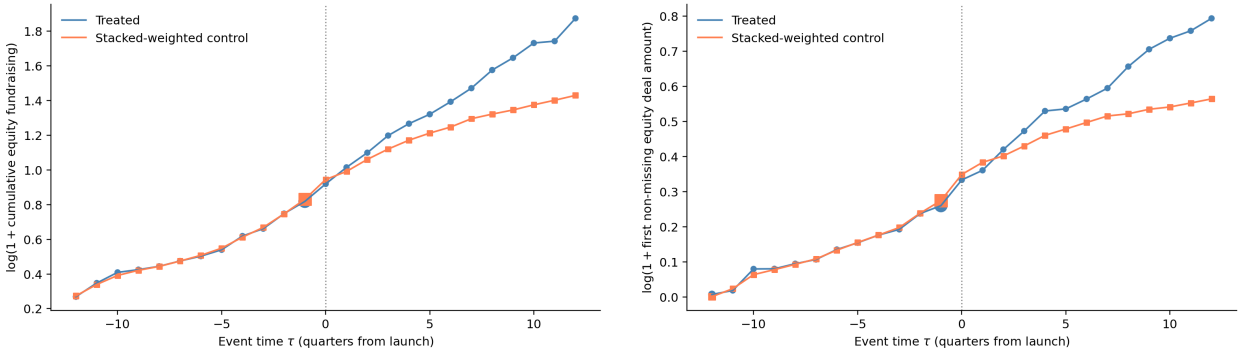


(A) Equity deal count in logarithms

(B) Equity amount (total recorded)

Notes: Panel (A) plots $\log(1 + x)$ of the cumulative number of equity rounds. Panel (B) plots $\log(1 + x)$ of cumulative recorded equity capital in real 2017 USD, with unreported amounts set to zero. Both panels follow Figure 3.

Figure A5: Pre-launch fit: alternative equity amount samples

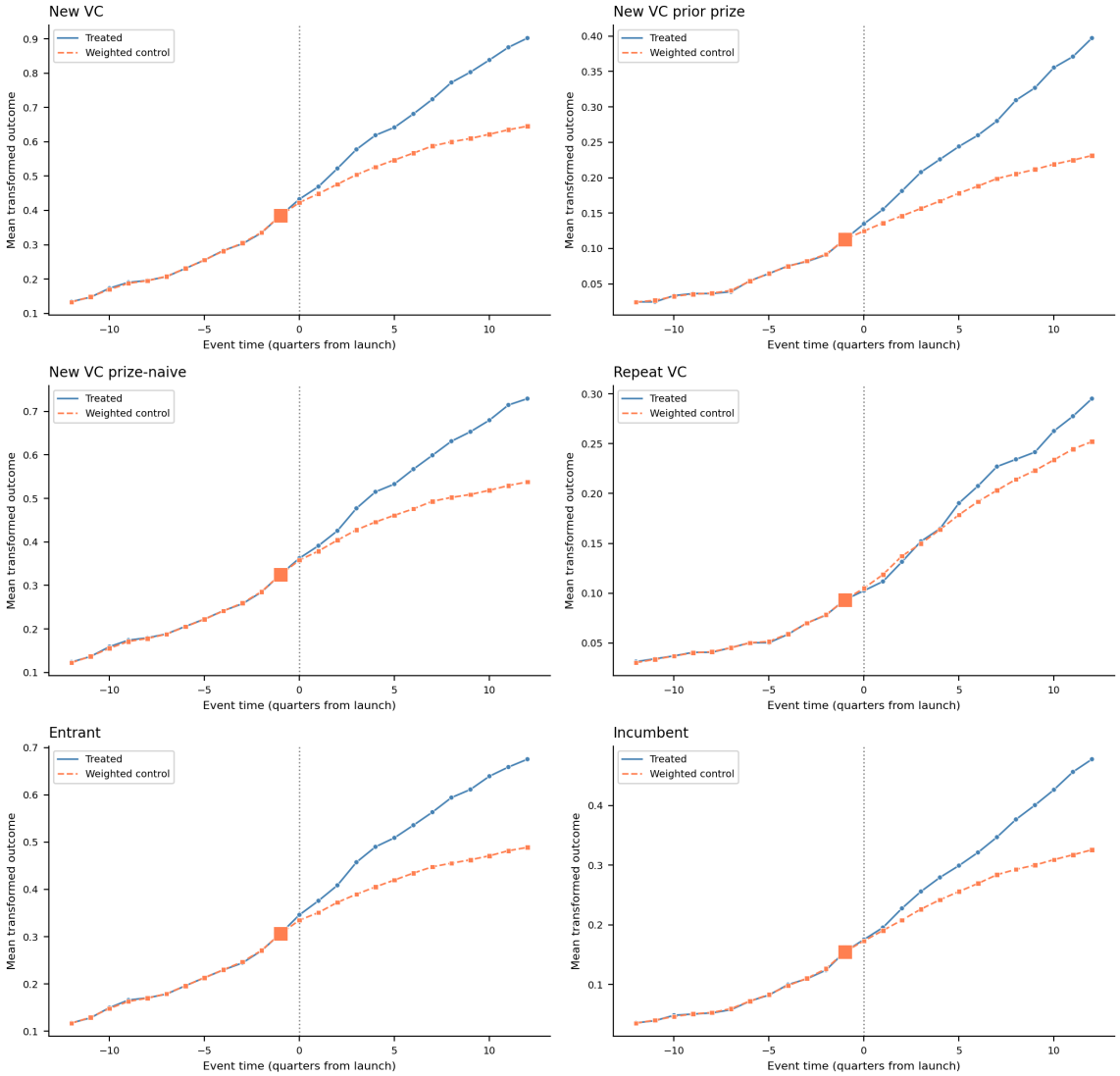


(A) Cumulative equity amount (complete-information sample)

(B) First recorded equity amount

Notes: Panel (A) restricts the sample to units with at least one in-window equity round and no in-window round without a reported amount; the outcome is $\log(1 + x)$ of cumulative equity capital. Panel (B) reports equity amount (first recorded), defined in Table A18. Construction follows Figure 3.

Figure A6: Pre-launch fit: new and prior investors

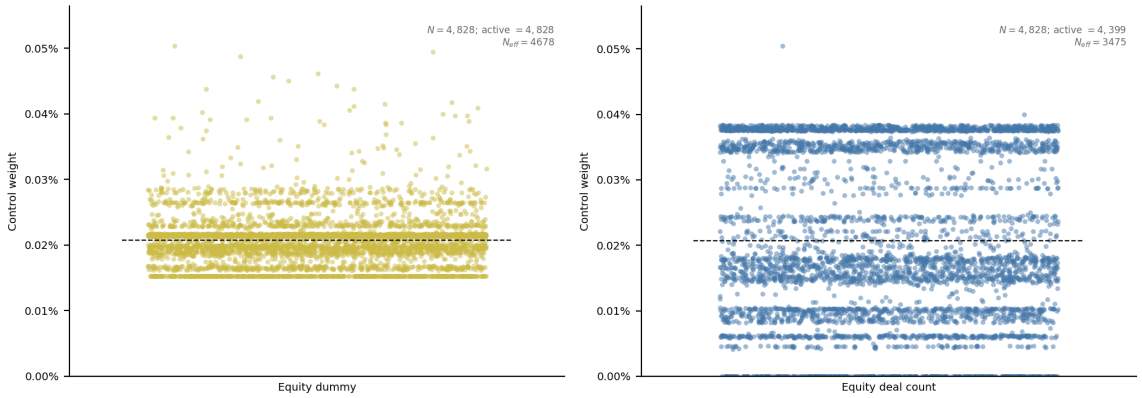


Notes: The six panels plot $\log(1 + x)$ of investor counts. The top four show VCs new to the firm, new to the firm but prior XPRIZE investors, new to XPRIZE, and prior firm investors (Panels A and B of Table 10). The bottom two show entrant and incumbent VCs new to the firm (Panel C): an entrant has no earlier investment in a firm tagged with any of the prize's ten leading participant keywords; an incumbent has at least one. Construction follows Figure 3.

B.3 Control-weight support

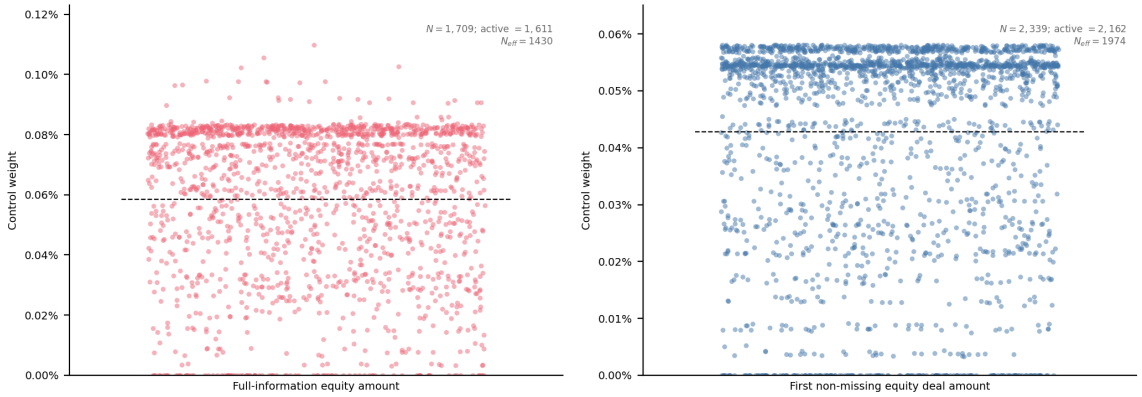
Figure A7 plots the SDID weight of every control unit. For the equity dummy, all 4,828 controls receive positive weight and the effective number of controls, $(\sum_i \widehat{\omega}_i^2)^{-1}$, is 4,678. For the equity deal count, 4,399 controls receive positive weight and the effective number is 3,475. The estimates therefore do not rest on a small set of controls. Figure A8 reports the same diagnostic for the amount samples. Figures **??** and **??** report it prize by prize.

Figure A7: SDID control weights



Notes: Each point is one control firm. The first panel reports weights for the equity dummy; the second reports weights for the equity deal count. The vertical axis reports the SDID unit weight. The dashed line marks the uniform weight $1/N_0$. N is the number of controls, "active" the number with positive weight, and N_{eff} the inverse of the sum of squared weights.

Figure A8: SDID control weights: equity amount samples



Notes: Construction follows Figure A7 for the samples of Figure A5.

C Robustness

This appendix reports the robustness tests discussed in Sections 4 to 7. Unless stated otherwise, every table uses the estimator, event window, outcomes, and inference of Table 2. Equity deal amount is $\log(1 + x)$ of cumulative recorded equity capital, as in the total-recorded measure of Appendix D.

C.1 Patent-class control pool

Table A8 replaces the keyword-based control pool with firms that share at least one Cooperative Patent Classification (CPC) subclass with the prize's participants. The pool

contains 4,349 control units. The effects are 8.8 percentage points and 0.516 rounds. Recorded equity capital rises by 0.179 log points. All estimates are significant at the 1% level.

Table A8: Equity fundraising with a patent-class control pool

Estimate	Equity dummy	Equity deal count	Equity deal amount	Non-dilutive dummy
CPC pre-launch control				
ATT	0.088***	0.516***	0.179***	0.014***
	(0.007)	(0.053)	(0.033)	(0.004)
N treated firms	383	383	383	383
N treated deals	823	823	823	307
N control firm	4,349	4,349	4,349	4,349

Notes: Controls share at least one of the twenty CPC subclasses most frequent among the prize's participants and record a deal in the five years before launch. Firm CPC subclasses come from PitchBook patent fields or, where these are missing, from matched USPTO patents. The non-dilutive dummy equals one if the firm raises a grant, debt, or product-crowdfunding deal in the quarter; its treated-deal row counts non-dilutive deals. N control firms are control firms. Other definitions follow Table 2.

C.2 Pre-launch fundraising

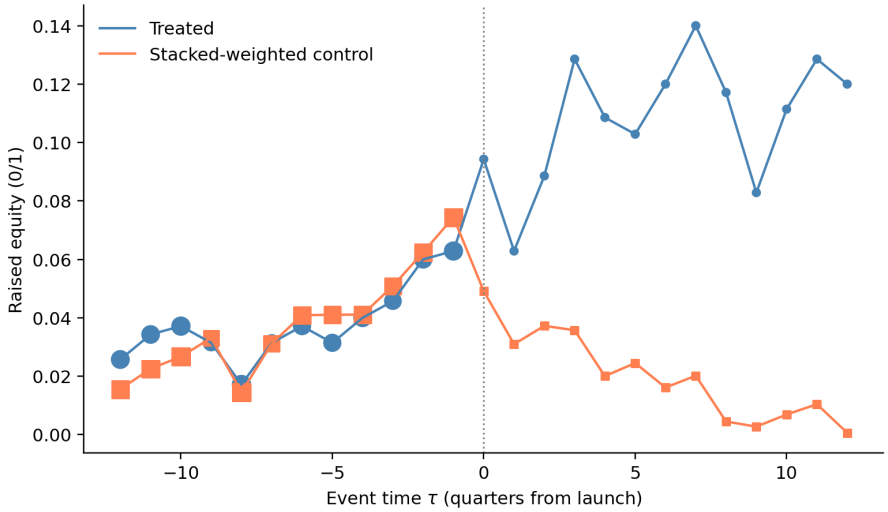
Figure 3 shows a rise in equity incidence in the two quarters before launch. If firms join a prize because they are about to raise capital, this rise would signal selection. We randomly remove half, then 70%, of the participant units with exactly one equity round in quarters $\tau = -2$ and $\tau = -1$ and re-estimate the weights. Such units have pre-launch equity, so the removal leaves the extensive margin unchanged. After removing half, 350 participant units remain (Table A9). The effects are 8.8 percentage points and 0.353 rounds, against 7.7 points and 0.338 rounds in Table 2. On the intensive margin, they are 4.7 points and 0.335 rounds, against 3.5 points and 0.307 rounds in Table 3. After removing 70%, 342 units remain (Table A10). The effects are 8.8 points and 0.349 rounds, and 4.3 points and 0.320 rounds on the intensive margin. Figures A9 and A10 show the pre-launch fit after each removal. Participant equity incidence at $\tau = -1$ falls from 11.0% in the baseline to 6.3% and 4.4%.

Table A9: Equity fundraising after removing half of the pre-launch fundraisers

Estimate	Equity dummy	Equity deal count	Equity deal amount	First recorded deal amount
Baseline sample				
ATT	0.088***	0.353***	0.086**	0.108**
	(0.006)	(0.047)	(0.037)	(0.052)
N treated firms	350	350	350	159
N treated deals	717	717	717	159

Notes: Half of the participant units with exactly one equity round in $\tau \in \{-2, -1\}$ are removed at random before the SDID weights are re-estimated. The "Baseline sample" panel pools all remaining participant units. Margins follow Table 3. First recorded deal amount is the equity amount (first recorded) of Table A18. Other definitions follow Table 2.

Figure A9: Pre-launch fit after removing half of the pre-launch fundraisers



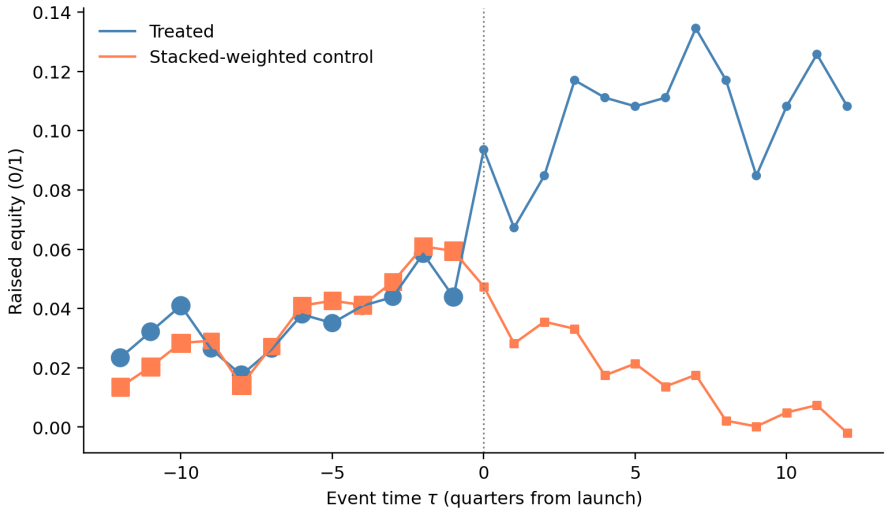
Notes: The outcome is the equity dummy for the pooled sample of Table A9. Construction follows Figure 3.

Table A10: Equity fundraising after removing 70% of the pre-launch fundraisers

Estimate	Equity dummy	Equity deal count	Equity deal amount	First recorded deal amount
Baseline sample				
ATT	0.088***	0.349***	0.085**	0.124**
	(0.006)	(0.045)	(0.036)	(0.050)
N treated firms	342	342	342	153
N treated deals	679	679	679	153

Notes: Seventy per cent of the participant units with exactly one equity round in $\tau \in \{-2, -1\}$ are removed at random. Other definitions follow Table A9.

Figure A10: Pre-launch fit after removing 70% of the pre-launch fundraisers



Notes: The outcome is the equity dummy for the pooled sample of Table A10. Construction follows Figure 3.

C.3 Award recipients

Table A11 removes award recipients from the participant sample while keeping the controls unchanged. Excluding winners leaves 368 participant units and effects of 7.8 percentage points and 0.344 rounds. Excluding winners and finalists leaves 309 units and effects of 8.3 points and 0.420 rounds. Both amount measures also remain positive and significant. Prize money and late-stage publicity therefore do not explain the baseline effect.

Table A11: Equity fundraising excluding award recipients

Estimate	Equity dummy	Equity deal count	Equity deal amount	First recorded deal amount
Excluding winners				
ATT	0.078***	0.344***	0.098**	0.113**
	(0.006)	(0.042)	(0.042)	(0.052)
N treated firms	368	368	368	176
N treated deals	787	787	787	176
Excluding winners and finalists				
ATT	0.083***	0.420***	0.146***	0.117***
	(0.008)	(0.058)	(0.043)	(0.044)
N treated firms	309	309	309	160
N treated deals	725	725	725	160

Notes: The first panel removes participant units that won a prize. The second removes winners and finalists. First recorded deal amount is the equity amount (first recorded) of Table A18. Other definitions follow Table 2.

C.4 Prize-size standardisation

Table 7 in Section 5.2 aggregates prize-level SDID estimates with the three cohort weights of Equation (6). Table A12 shows how the weights move. XPRIZE Carbon Removal carries 32.1% of the standard weight but only 2.3% of the inverse-purse weight. XPRIZE Rapid Covid Testing, with an expected prize of USD 0.027 million, carries 51.5% of the inverse-expected-prize weight.

Table A12: Prize weights under the three standardisations

Prize	Cohort-size weight (%)	Inverse purse weight (%)	Inverse expected prize weight (%)	Expected prize (USD m)	Total purse (USD m)
Avatar	6.32%	4.18%	3.72%	0.109	9.78
Adult Literacy	1.92%	1.73%	0.62%	0.200	7.19
Global Learning	0.82%	0.34%	0.04%	1.197	15.56
AI	9.07%	11.52%	8.33%	0.070	5.09
Carbon	5.49%	1.73%	0.51%	0.685	20.55
Sensing	4.67%	12.51%	2.85%	0.105	2.41
Pandemic Response	1.37%	18.72%	19.64%	0.004	0.47
Automotive	0.55%	0.31%	0.08%	0.418	11.29
Tricorder	1.37%	0.83%	0.11%	0.825	10.73
Ocean Discovery	0.82%	0.74%	0.18%	0.288	7.19
Water Abundance	0.55%	2.00%	0.14%	0.254	1.78
Ocean Health	1.10%	3.37%	0.63%	0.111	2.11
Women's Safety	1.37%	8.73%	1.81%	0.048	1.02
Carbon Removal	32.14%	2.29%	7.98%	0.258	90.79
Feed the Next Billion	7.42%	3.37%	1.07%	0.445	14.24
Rainforest	2.20%	1.48%	0.59%	0.240	9.62
Rapid COVID Testing	21.70%	24.65%	51.48%	0.027	5.70
Rapid Reskilling	1.10%	1.50%	0.21%	0.339	4.75

Notes: Each column reports the share of total weight assigned to a prize cohort under the weighting of Table 7. Weights sum to 100% across the 18 prizes. Expected prize equals the headline purse divided by the number of participating teams. Monetary values are in real 2017 USD millions.

Table A13 compares prize purses with the funding that participants record around launch. Averaged across the 18 prizes, the participants of a prize record USD 47.71 million of equity funding in the twelve quarters before launch and USD 204.16 million in the twelve quarters from launch. Average grant funding rises from USD 13.67 million to USD 18.38 million. The ratio of equity and grant funding to the total purse averages 8.22 before launch and 15.75 after. These averages rest on a few prizes. Equity funding rises from USD 69.28 million to USD 1,336.83 million in XPRIZE Carbon Removal and from USD 283.79 million to USD 1,655.93 million in XPRIZE Feed the Next Billion. Two prizes record no equity funding in either window. In eight of the 18 prizes, funding after launch falls below the total purse. The comparison is descriptive. It has no control group, and unreported deal amounts count as zero.

Table A13: Prize purses and participant funding around launch

Prize	Total purse (USD m)	Expected prize (USD m)	Total equity funding (USD m)		Total grant funding (USD m)		Funding-to prize ratio	
			Pre	Post	Pre	Post	Pre	Post
Avatar	9.78	0.109	12.02	20.39	2.06	0.14	1.44	2.10
Adult Literacy	7.19	0.200	0.00	0.27	0.24	0.98	0.03	0.17
Global Learning	15.56	1.197	2.00	4.11	0.00	0.00	0.13	0.26
AI	5.09	0.070	90.18	85.13	4.00	6.98	18.50	18.10
Carbon	20.55	0.685	27.33	24.50	1.58	17.63	1.41	2.05
Sensing	2.41	0.105	5.70	23.89	0.38	4.67	2.52	11.83
Pandemic Response	0.47	0.004	0.02	0.27	0.00	0.01	0.04	0.59
Automotive	11.29	0.418	0.00	0.00	0.00	0.00	0.00	0.00
Tricorder	10.73	0.825	0.00	17.42	1.82	0.91	0.17	1.71
Ocean Discovery	7.19	0.288	0.00	0.00	0.64	1.08	0.09	0.15
Water Abundance	1.78	0.254	5.28	0.00	2.04	0.00	4.11	0.00
Ocean Health	2.11	0.111	0.00	0.39	0.00	0.00	0.00	0.19
Women's Safety	1.02	0.048	4.07	6.58	0.00	0.15	4.00	6.61
Carbon Removal	90.79	0.258	69.28	1336.83	34.58	162.70	1.14	16.52
Feed the Next Billion	14.24	0.445	283.79	1655.93	27.07	4.06	21.83	116.59
Rainforest	9.62	0.240	12.90	56.33	0.13	0.29	1.35	5.89
Rapid COVID Testing	5.70	0.027	340.57	441.44	169.52	131.23	89.56	100.55
Rapid Reskilling	4.75	0.339	5.57	1.35	2.07	0.00	1.61	0.29
Average across prizes	12.24	0.312	47.71	204.16	13.67	18.38	8.22	15.75

Notes: Each row is a prize cohort in Table A12. The total purse is the headline purse; the expected prize is that purse divided by the number of participating teams. Equity and grant funding are the sums of recorded deal amounts across the prize's matched participant firms in the twelve quarters before launch ($\tau = -12$ to -1) and the twelve quarters from launch ($\tau = 0$ to 11). The funding-to-prize ratio divides the sum of equity and grant funding in each window by the total purse. The last row reports the unweighted mean of each column across the 18 prizes. All amounts are in real 2017 USD millions. Unreported deal amounts count as zero; grant funding excludes XPRIZE award payments. Some firms have shorter observed histories; amounts outside their observed quarters are not extrapolated.

C.5 Carbon Removal exclusion

XPRIZE Carbon Removal is the largest cohort and has the largest purse. Table A14 removes its participants and controls. The remaining 266 participant units and 3,441 controls give effects of 6.2 percentage points and 0.256 rounds, both significant at the 1% level. The recorded equity amount and non-dilutive incidence estimates are positive but not significant.

Table A14: Equity fundraising excluding XPRIZE Carbon Removal

Estimate	Equity dummy	Equity deal count
No-carbon control		
ATT	0.062***	0.256***
	(0.008)	(0.052)
N treated firms	266	266
N treated deals	506	506
N control firm	3,441	3,441

Notes: Participants and controls of XPRIZE Carbon Removal are removed. Other definitions follow Table A8.

C.6 Prize-level estimates

Table A15 estimates the effect separately for each prize cohort. The equity-dummy effect is positive and significant in all eight cohorts. It ranges from 3.4 percentage points for XPRIZE Rapid Covid Testing to 10.5 points for XPRIZE Carbon Removal. The deal-count effect is significant for IBM Watson AI XPRIZE, XPRIZE Carbon Removal, and XPRIZE Feed the Next Billion. No single prize drives the baseline effect.

Table A15: Equity fundraising by prize cohort

Prize	Equity dummy	Equity deal count
ANA Avatar XPRIZE	0.071***	0.061
	(0.027)	(0.180)
IBM Watson AI XPRIZE	0.089***	0.457***
	(0.018)	(0.154)
NRG Cosia Carbon XPRIZE	0.077**	0.301
	(0.035)	(0.302)
Nokia Sensing XCHALLENGE	0.052***	0.087
	(0.020)	(0.132)
Other	0.046***	0.067
	(0.013)	(0.080)
XPRIZE Carbon Removal	0.105***	0.497***
	(0.010)	(0.081)
XPRIZE Feed the Next Billion	0.101***	0.897***
	(0.036)	(0.240)
XPRIZE Rapid Covid Testing	0.034**	0.087
	(0.014)	(0.077)

Notes: Each row applies the SDID estimator to one prize cohort and its control pool. "Other" pools prizes with fewer than ten matched participants, each aligned on its own launch date. Other definitions follow Table 2.

C.7 Technological proximity within the control pool

This test asks whether the treatment effect depends on how close controls are to the prize's technology. It keeps the participants and the baseline control pool of Table 2 and splits the pool by technological proximity. It therefore differs from the distant ring of Section 5.3, which adds firms from outside the pool. Every control in the pool shares at least one of the participants' twenty leading keywords. The split measures how much more it shares.

Proximity is the keyword score of Equation (5), $\pi_{ip} = \sum_{w \in K_i} n_p(w) / \sum_{j \in \mathcal{T}_p} |K_j|$. The denominator counts every keyword tag of the prize's participants. The numerator counts the tags that fall on keywords control i also carries. π_{ip} is therefore the share of participant keyword tags matched by the control and lies between zero and one. A keyword carried by many participants raises proximity more than a keyword carried by one. For example, if 30 of the participants' 100 keyword tags are "carbon capture", a control tagged only with "carbon capture" has $\pi_{ip} = 0.30$. The score uses all participant keywords, not only the twenty that define the pool. For the "Other" cohort, $n_p(w)$ counts the participants of all its prizes.

Within each prize, we split the controls at the median of π_{ip}. Controls at or above the median form the close group, and controls below it the distant group. Ties at the median enter the close group, so the groups are unequal: 3,633 and 1,195 control units. Each group is compared with all 383 participant units, and the SDID weights are re-estimated within each group. The difference row re-estimates both effects on the same bootstrap draws.

Table A16 reports the results. The equity-dummy effect is 7.7 percentage points against close controls and 7.6 against distant ones. The deal-count effect is 0.321 rounds against close controls and 0.378 against distant ones. The count difference of 0.056 is significant at the 5% level. Participants therefore gain slightly more relative to controls that are further from their technology, consistent with the spillover of Section 5.3.

Table A16: Equity fundraising by technological proximity of controls

Estimate	Equity dummy	Equity deal count	Equity deal amount
Technologically close controls			
ATT	0.077***	0.321***	0.081**
	(0.004)	(0.048)	(0.041)
Technologically distant controls			
ATT	0.076***	0.378***	0.118***
	(0.007)	(0.051)	(0.034)
Distant controls − close controls			
Difference	-0.001	0.056**	0.037*
	(0.004)	(0.025)	(0.020)
N treated units		383	
N control units (baseline / close / distant)		4,828 / 3,633 / 1,195	

Notes: Close controls have keyword proximity π_{ip} (Equation (5)) at or above the within-prize median of the control pool; distant controls have proximity below it. Both groups belong to the baseline control pool and differ from the distant ring of Table 8. The difference row reports the effect for distant controls less the effect for close controls, with paired-bootstrap standard errors. Other definitions follow Table 2.

C.8 VC and non-VC equity

Table A17 narrows the definition of equity. Counting only VC rounds, the effects are 6.5 percentage points and 0.165 rounds for 202 participant units. Counting VC and non-VC rounds with a recorded amount, they are 9.2 points and 0.222 rounds for 268 units. Restricting to non-VC rounds yields an incidence effect of 7.6 percentage points and a deal-count effect of -0.014 rounds for 182 units; only the incidence effect is statistically distinguishable from zero. The VC-only effect persists when the outcome counts only rounds involving a VC investor.

Table A17: Equity fundraising by type of equity round

Estimate	Equity dummy	Equity deal count
VC-only equity rounds		
ATT	0.065***	0.165***
	(0.007)	(0.057)
N treated firms	202	202
N treated deals	425	425
VCs and non-VCs with a recorded equity-deal amount		
ATT	0.092***	0.222***
	(0.007)	(0.046)
N treated firms	268	268
N treated deals	670	670
Non-VC-only equity rounds		
ATT	0.076***	-0.014
	(0.007)	(0.048)
N treated firms	182	182
N treated deals	300	300

Notes: The first panel counts only rounds with a VC investor as equity. The second also counts rounds with a non-VC investor who has a recorded equity-deal amount in the investor's deal history. The third counts rounds with a listed non-VC investor but no listed VC investor, whether or not a deal amount is recorded; firm–prize units with no such round in the event window are excluded, including those with only VC rounds. Each estimate uses 50 unit-bootstrap draws. Other definitions follow Table 2.

D Equity amounts and valuations

The main text measures equity fundraising by its incidence and its count. This appendix reports the effect on amounts raised and on valuations. Both outcomes rest on incomplete data. PitchBook reports no amount for 402 of the 894 in-window equity deals of matched participants (45.0%; Appendix A.2). The missing amounts are not randomly distributed across firms and mostly affect younger and smaller firms. No single measure handles this gap, so we use two measures with complementary strengths. *Equity amount (total recorded)* is $\log(1 + x)$ of cumulative recorded equity capital, with unreported amounts set to zero. It uses every participant unit but understates capital when amounts are missing. *Equity amount (first recorded)* is $\log(1 + x)$ of the first equity round with a reported amount, from the quarter of that round onwards. It imputes no zeros but retains only the 182 participant units whose first reported round falls within the event window.

D.1 Equity amounts

Table A18 shows that both measures point to more capital. Total recorded equity capital rises by 0.093 log points, or 9.7%, significant at the 1% level. The first recorded round is 0.107 log points, or 11.3%, larger, significant at the 5% level. The additional rounds of Section 4 therefore bring additional capital. Two further estimates point the same way. Counting only VC rounds, as in the VC-only sample of Table A17, recorded equity capital rises by 10.6%, significant at the 10% level. Grant capital rises by 5.8%, significant at the 1% level, and its difference with the equity-capital effect is not significant.

Table A18: Equity amounts raised

Estimate	Equity amount (total recorded)	Equity amount (first recorded)
Baseline sample		
ATT	0.093^{***}	0.107^{**}
	(0.030)	(0.044)
N treated firms	383	182
N treated deals	823	182

Notes: Entries are stacked SDID estimates of the treatment effect, with standard errors from 50 unit-bootstrap replications in parentheses. Outcomes and support rows follow Table 2. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Equity amount (total recorded) is $\log(1+x)$ of the cumulative recorded equity capital in real 2017 USD, with unreported deal amounts set to zero. Equity amount (first recorded) is $\log(1+x)$ of the amount of the first equity round with a reported amount, set from the quarter of that round onwards; it retains only units whose first reported round falls within the event window. A coefficient β corresponds to a proportional change of $\exp(\beta) - 1$.

D.2 Valuation

Table A19 reports the effect on the logarithm of the post-money valuation, which PitchBook records for few firms. The intensive-margin estimate is 0.354 log points, or 42.5%, for 58 participant units, significant at the 5% level. The extensive-margin estimate is 0.604 log points, or 82.9%, for 16 units, significant only at the 10% level. These small samples limit precision, particularly on the extensive margin; the estimates are consistent with higher recorded valuations alongside the fundraising effects.

Table A19: Valuation on the intensive and extensive margins

Estimate	Valuation
At least one pre-prize equity round (intensive margin)	
ATT	0.354**
	(0.140)
N treated firms	58
N treated deals	120
No pre-prize equity fundraising (extensive margin)	
ATT	0.604*
	(0.352)
N treated firms	16
N treated deals	19

Notes: The outcome is the logarithm of the recorded post-money valuation in real 2017 USD. The sample retains units with an observed valuation, and deal support counts participant deals with a recorded valuation. Margins follow Table 3. Other definitions follow Table 2.

E Investor Composition

E.1 Technology-space entrants: twenty keywords

Table A20 repeats Panel C of Table 10 with the twenty leading participant keywords of each prize. A VC is new to the technology space if it has never invested in a firm tagged with any of these keywords. The broader definition classifies more VCs as incumbents. Incidence rises by 2.7 percentage points for entrants and 2.1 points for incumbents. The investor count rises by 0.101 and 0.069 log points. The differences, 0.6 points (0.004) and 0.031 (0.020), keep the sign of the ten-keyword results but are not significant.

Table A20: Technology-space entrants and incumbents: twenty keywords

Investor type	Investor dummy	Investor count
Entrant_{20}	0.027***	0.101***
	(0.003)	(0.016)
Incumbent_{20}	0.021***	0.069***
	(0.003)	(0.016)
$\text{Entrant}_{20} - \text{Incumbent}_{20}$	0.006	0.031
	(0.004)	(0.020)
N treated firms	383	383
N treated equity deals	823	823

Notes: Entrant_{20} is a VC new to the firm with no earlier investment in a firm tagged with any of the prize's twenty leading participant keywords; Incumbent_{20} has at least one such investment. Invested amount is $\log(1 + x)$ of cumulative capital allocated by the investor type, dividing each syndicated deal equally across its listed investors. Other definitions follow Table 10.

E.2 Investor geography

Table A21 splits new VCs by headquarters. Across all participants, the number of US and non-US VCs rises by similar amounts, 0.087 and 0.080 log points. The split by firm location reveals home bias. Among US participants, the number of US VCs rises by 0.120 log points and that of non-US VCs by 0.033, a difference of 0.086 (0.030). Among non-US participants, the ranking reverses: non-US VCs rise by 0.122 and US VCs by 0.055, a difference of −0.068 (0.026). Both differences are significant at the 1% level. The prize signal thus attracts mainly investors located near the firm.

Table A21: New investors by geography

Firm sample and investor type	Investor count	Invested amount
All firms: US VC	0.087***	0.049**
	(0.021)	(0.025)
All firms: Non-US VC	0.080***	0.065***
	(0.014)	(0.022)
All firms: US VC − Non-US VC	0.007	-0.016
	(0.019)	(0.020)
US firms: US VC	0.120***	0.057
	(0.036)	(0.039)
US firms: Non-US VC	0.033*	0.039*
	(0.017)	(0.023)
US firms: US VC − Non-US VC	0.086***	0.018
	(0.030)	(0.032)
Non-US firms: US VC	0.055***	0.042**
	(0.021)	(0.021)
Non-US firms: Non-US VC	0.122***	0.091**
	(0.026)	(0.042)
Non-US firms: US VC − Non-US VC	-0.068***	-0.049
	(0.026)	(0.034)
N treated firm (all / US / non-US firms)	383 / 194 / 189	383 / 194 / 189
N treated deals (all / US / non-US firms)	1,130 / 594 / 536	741 / 433 / 308

Notes: Investor count is $\log(1 + x)$ of the cumulative number of distinct VCs new to the firm and headquartered in the stated location. Invested amount is $\log(1 + x)$ of the cumulative capital they allocate. Firm samples split participant units by firm headquarters. Difference rows use paired-bootstrap standard errors. Other definitions follow Table 10.